

Physics-augmented neural networks and projection-based reduced-order models for anisotropic hyperelasticity

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Abstract

Approximating the effective anisotropic hyperelastic response of a representative volume element (RVE) requires constitutive expressiveness, mechanical consistency, and computational efficiency. Non-intrusive neural surrogates provide direct constitutive evaluations, whereas intrusive projection-based reduced-order models (PROMs) approximate the microscopic state and equilibrium equations. This work develops a polyconvex physics-augmented neural network (PANN) construction based on paired directional features with trainable orientations, stretch powers, and determinant exponents. The features are convex in independent deformation minors and are composed with convex, componentwise nondecreasing cores realized as an input-convex neural network (ICNN) or an input-convex Kolmogorov–Arnold network (ICKAN). Their isotropic reference derivatives enable affine-determinant stress normalization without compromising polyconvexity. The complete energies satisfy objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and prescribed growth conditions by construction; global nonnegative energy is assessed through a separate sufficient bound for the trained parameters. Complementary intrusive models use empirical cubature method (ECM) hyperreduction on reduced meshes: an affine hyperreduced PROM (HPROM), a nonlinear-manifold HPROM–ANN with neural latent-coordinate closure and manifold-adaptive weights ECM (MAW–ECM), and a direct decoder variant without reduced equilibrium iterations. On a periodic cell with a rotated elliptical void, the selected polyconvex models attain approximately 0.65% held-out stress error. Closed-cycle tests distinguish constitutive consistency from prediction accuracy, exposing nonconservativity in direct stress regression and all three deployed HPROM variants. A tensile-coupon application illustrates structural deployment against a full-order FE² reference. The ICNN and ICKAN give displacement errors below 0.43%; the iterative HPROM–ANN gives 0.052% error and its implemented workflow is 2.23 times faster than the optimized affine HPROM under the reported configurations. These results support the proposed constitutive construction and identify complementary accuracy–cost choices without equating successful structural convergence with a constitutive guarantee.

Keywords: computational homogenization; hyperelasticity; physics-augmented neural networks; polyconvexity; projection-based reduced-order models; empirical cubature.

1 Introduction

The effective response of a heterogeneous material emerges from the deformation of its constituents and their interactions with the microstructure. At finite strains, this interaction can produce anisotropy and coupling between loading modes that are difficult to capture with a predetermined macroscopic constitutive law. Computational homogenization addresses this difficulty by resolving a representative volume element (RVE): an imposed deformation drives a microscopic boundary-value problem, from which effective stresses and constitutive derivatives are obtained [1]. Rather than prescribing the effective law in advance, the RVE supplies it through the solution of the underlying mechanical problem.

This flexibility comes at a computational cost. Characterizing the material over a range of deformations already requires many nonlinear finite element solves; using the same RVE within a structural calculation multiplies those solves across macroscopic integration points and load increments. Much of this repeated work concerns closely related microscopic states. Surrogate modeling seeks to exploit that relationship by computing representative solutions offline and reusing their information in less expensive online evaluations.

The objective is therefore not only to reduce execution time and the resources required to store and process microscopic fields, but to do so without losing the constitutive behavior that made the RVE description useful.

There are two complementary ways to pursue this objective. Non-intrusive constitutive surrogates learn the effective response from RVE input–output data and dispense with the microscopic solve online. Intrusive projection-based reduced-order models (PROMs) instead compress the microscopic displacement field and retain projected equilibrium equations, with hyperreduction limiting the local contributions needed to evaluate them. The former offers a direct constitutive evaluation; the latter retains access to an approximate microscopic state and the given constituent law. Mechanical knowledge and learning can enter either route: physics-augmented neural networks (PANNs) embed constitutive restrictions in the learned response, while artificial neural networks (ANNs) can enrich a PROM’s displacement representation.

This work develops a polyconvex PANN for the effective anisotropic hyperelastic response of an RVE. The aim is to make the learned energy sufficiently expressive while preserving its mechanical structure, including an exactly stress-free reference state. PROM-based surrogates of the same RVE provide complementary accuracy–cost comparisons, and a tensile structural example examines how the constitutive approximations behave when deployed beyond isolated material evaluations. The constitutive construction is the central contribution; structural deployment supplies a further test of its usefulness.

1.1 Physics-augmented neural networks

The attraction of direct constitutive learning is that the expensive microscopic solver is needed to generate data, but not to evaluate the trained material model. Neural homogenization has demonstrated this separation [2], making the systematic generation of representative microscopic responses an important part of the offline workflow [3]. Once this separation is made, however, the inputs and architecture of the surrogate must carry the information that would otherwise be supplied by the microscopic problem.

For materials with memory, this information extends beyond the current deformation. Loading history can be retained through learned history-to-response maps [4] or through internal descriptors, such as the maximum directional stretches used in polymer models [5]. The data-generation burden also grows when several interacting scales must be represented; staggered training of successive two-scale surrogates provides one way to manage that burden [6]. Here we consider history-independent hyperelasticity. Removing the need to represent memory simplifies the state description, but introduces a precise constitutive requirement: the stress response must be compatible with a stored-energy function.

Learning that function, rather than fitting each stress component independently, makes the stress work conservative and gives a major-symmetric constitutive tangent by differentiation. This is a central motivation for the mechanics-informed energy formulation of As’ad et al. [7]. Expressing the energy through objective deformation measures then removes dependence on a superposed rigid rotation. In this way, PANNs shift part of the burden from learning mechanical identities from data to satisfying them through the representation itself [8, 9]. Prediction accuracy remains to be established, but potential structure and objectivity need not depend on how densely the training domain has been sampled.

Preserving a potential does not, by itself, control its curvature. At finite strains, that control must also be imposed in appropriate variables. Convexity in Green–Lagrange strain, as used in Ref. [7], is attractive because stress and tangent follow directly from the learned energy. Its implication for curvature with respect to the deformation gradient nevertheless contains a stress-dependent term [10], so it is not a global polyconvexity guarantee. Conversely, enforcing a positive-semidefinite matrix through a lower-triangular factorization [11] controls a matrix property, but does not by itself supply the integrability needed to recover a scalar energy. The modeling task is to preserve both the potential and the desired finite-strain curvature condition within one construction.

Polyconvexity provides a route to that combination. It represents the energy as a convex function of deformation minors, yielding a sufficient condition for rank-one convexity [12]. For the two-dimensional model considered here, the independent arguments are the deformation gradient and its determinant. This framework also underpins existence results when the additional coercivity and admissibility hypotheses are met [13]. We use it as a sufficient constitutive restriction, not as a necessary property of every homogenized material or a guarantee of structural uniqueness. The remaining challenge is representational: the restrictions that provide a guarantee must still leave enough freedom to describe the anisotropic RVE response.

1.2 Projection-based reduced-order models

An alternative to learning the effective energy is to retain the microscopic problem and reduce the space in which it is solved. In a conventional PROM, solution snapshots supply a reduced-order basis, usually through proper orthogonal decomposition (POD), and projected equilibrium determines the coefficients of

the displacement field in that basis. This allows information from previous RVE solves to be reused without replacing the constituent law, as demonstrated in reduced finite-strain homogenization [14, 15]. Retaining a field representation also supports multiscale settings in which reduced periodic substructures must exchange kinematic information [16]. For the present RVE, the question is how much of this field information can be retained at a small online cost.

The first limitation is the approximation space. A family of solutions may depend on only a few loading parameters and still be poorly represented by a small global linear basis. The Kolmogorov n -width makes this distinction precise: if the best worst-case error over all n -dimensional linear spaces decays slowly, adding modes remains expensive even when the solution family has a low-dimensional parametrization [17, 18]. Local bases address this difficulty by distributing the approximation among regions of the solution set [19]. Nonlinear manifolds pursue the same objective without restricting each representation to an affine space, using, for example, autoencoder mappings [20] or quadratic corrections [21].

This motivates retaining a rich POD representation while reducing the number of coefficients that must be determined independently. The PROM-ANN formulation [17] achieves this by solving for a small set of primary coordinates and using an ANN to reconstruct the secondary ones.¹ The displacement field can thus retain the contribution of many POD modes without introducing an equilibrium unknown for each one. We adopt this primary/secondary representation with an ANN closure for the RVE, seeking to retain the accuracy of the richer displacement space at the cost of a smaller nonlinear system.

Reducing the unknowns does not eliminate a second cost: evaluating the microscopic contributions entering the projected equations. Even a very small reduced system can remain expensive if its residual and tangent require constitutive evaluations throughout the original mesh. Hardware acceleration can lower the cost of those evaluations, as in GPU-assisted reduced homogenization [22]; hyperreduction instead reduces how many contributions must be evaluated. It is this second approximation that turns a PROM into a hyperreduced projection-based reduced-order model (HPROM).

Two broad families approach this task from different directions [23, 24]. Approximate-then-project methods, notably discrete empirical interpolation (DEIM) [25], reconstruct a high-dimensional nonlinear quantity from selected samples before projecting its approximation. For a mechanical model, however, accurate reconstruction of that quantity does not automatically preserve the variational structure of the reduced equations. Project-then-approximate methods target the reduced quantities directly, often by replacing full assembly with a weighted sum of selected local contributions. Optimized cubature [26], energy-conserving sampling and weighting (ECSW) [27, 28], and the empirical cubature method (ECM) [29] follow this route. The structure-preservation results for ECSW in structural dynamics [28] illustrate the value of designing the approximation around the mechanical equations, although they do not constitute a stability guarantee for every sampling-and-weighting method. For the RVE considered here, this route also has a direct computational advantage: selected finite element contributions can be evaluated on a reduced mesh using the existing constituent law. We adopt ECM within this setting.

The original ECM construction [29] obtains a sparse integration rule by selecting spatial points and nonnegative weights that reproduce a training space of integrands. Its economy comes from replacing the many contributions in the full discretization by a much smaller weighted subset. The resulting rule nevertheless retains two restrictions: points are selected from a discrete candidate set, and one weight vector must serve the entire training space. Subsequent developments show that either restriction can be relaxed to seek greater compression. The continuous empirical cubature method (CECM) [30] takes an initially feasible discrete rule and removes points while readjusting both the positions and weights of those that remain. Here the extra freedom is spatial: the integration points are no longer confined to the original candidates.

For reduced models built from local spaces or nonlinear manifolds, the second restriction becomes especially relevant. Once different regions of the solution set have their own local bases, requiring one fixed weight vector to reproduce the conditions from all of them can impose an unnecessarily global integration problem. Subspace-adaptive weights ECM (SAW-ECM) [31] exploits this observation by retaining a common set of points while assigning different nonnegative weights to each local basis. As the local representation was refined in the examples of that study, the required common support decreased. The gain came from adapting how the selected points contribute, rather than from moving their locations.

Manifold-adaptive weights ECM (MAW-ECM) [32] carries this idea from a collection of local spaces to a continuously parameterized solution manifold: latent coordinates replace the local-basis index, and each retained spatial entity receives a weight field over the latent domain. The integration rule can thus follow the reduced state instead of reproducing the global linear span of all training conditions with one fixed vector. This is the connection to the nonlinear closure above. The MAW-ECM study illustrates it with an RBF-based reconstruction; the present work uses an ANN closure and neural weight predictors. The

¹The closure need not be neural: the latent-space framework of Ares De Parga et al. [18] also considers Gaussian-process regression (GPR) and radial-basis-function (RBF) closures.

adaptive-weight principle concerns the manifold and its integrands, not a particular regression architecture.

These developments motivate two complementary reduced evaluators for the present comparison: an affine HPROM with fixed ECM, and a nonlinear-manifold HPROM-ANN with adaptive weights. A direct mode, D-HPROM-ANN, additionally uses strain-informed coordinates to reconstruct the microscopic response without reduced equilibrium iterations. Comparing these modes examines how displacement compression, integration compression, and online equilibrium work affect accuracy and cost. Their constitutive outputs must still be assessed on their own terms: retaining the constituent law and positive cubature weights does not alone establish that every approximate effective stress derives from one energy.

1.3 Feature representation and position relative to prior work

The two surrogate routes therefore place the approximation in different parts of the problem. A PROM compresses the microscopic state while continuing to evaluate the given material law; an energy-based PANN must represent the effective law itself. For an anisotropic response, this requires more than a flexible scalar network. The inputs must express directional dependence while remaining compatible with objectivity and the desired convexity condition.

Structural tensors provide a way to encode material directions in objective scalar quantities and have enabled anisotropic polyconvex energies [33]. The directional description matters as much as the scalar response assigned to it: models of dispersed fibers, for example, account for a distribution of orientations rather than one preferred axis [34]. In a learned constitutive law, the corresponding feature map determines which distinctions between deformation states the network can use. Encoding directions is therefore part of the model, not merely a preprocessing step.

An ICNN provides a convex scalar core [35], but composing that core with nonlinear features requires additional care. A sufficient construction uses features convex in the independent deformation minors and a core that is also nondecreasing in each feature [8]. The latter condition links admissibility to expressiveness. If two states satisfy $\mathbf{z}(\mathbf{F}_a) \leq \mathbf{z}(\mathbf{F}_b)$ componentwise, a monotone core must satisfy $H(\mathbf{z}(\mathbf{F}_a)) \leq H(\mathbf{z}(\mathbf{F}_b))$. Increasing its width cannot reverse that ordering. When reference corrections and growth terms are present, the comparison concerns the part of the energy represented by the core, but the restriction remains. An unsuitable feature family can therefore prevent an otherwise admissible architecture from fitting the target response.

This is not merely a formal possibility. In the cubic-lattice example of Klein et al. [8], an invariant-based polyconvex model has difficulty representing shear, while a deformation-gradient-based formulation improves the fit. The latter uses data augmentation for approximate objectivity, whereas the invariant model is exactly objective and performs well for their transversely isotropic example. These results point to the representation, rather than polyconvexity alone, as a potential source of limited accuracy. They motivate enlarging the admissible feature family before concluding that a convex architecture is inadequate for a material.

Allowing the directional description to adapt during training is one way to gain that flexibility. Generalized-structure-tensor models learn aspects of the anisotropy [36], and objective triclinic polyconvex constructions show that a nontrivial material symmetry group need not be prescribed in advance [37]. Yet directional flexibility must also be reconciled with the reference state. A learned anisotropic energy can produce a nonzero reference stress, and cancelling that stress with an arbitrary strain correction can destroy polyconvexity. The compatible normalization and growth terms developed in Ref. [9] show why these corrections belong to the constitutive construction itself. This leads to the design question pursued here: can the directional features be made flexible while controlling their reference derivatives so that stress normalization remains admissible?

Once the features convey the required directional information, the scalar core must resolve how the energy varies with them. This flexibility can be placed in the coefficients of admissible energy terms, as in constitutive artificial neural networks (CANNs) [38], or in learned monotone energy derivatives [39]. For the present construction, an ICNN supplies a compatible core through weighted compositions of fixed convex activations. A spline-based Kolmogorov-Arnold network instead learns the shapes of its univariate edge functions, providing a complementary way to distribute that flexibility. Constitutive KAN formulations [40] motivate this alternative, but preserving the feature-composition proof requires more than using a scalar energy output: each edge must also respect the required convexity and monotonicity. Monotonic input-convex KANs (ICKANs) provide this connection [41]. We therefore assess the paired features with both cores and use analytically joined splines so that the admissibility argument also covers their tails. The comparison tests the scalar approximation while retaining the same type of mechanically constrained feature representation.

1.4 Present contribution

We address the feature and normalization requirements together through paired directional features with trainable orientations, stretch powers, and determinant exponents constrained by explicit joint-convexity

bounds. Pairing makes each feature’s strain derivative at the undeformed state proportional to the identity tensor, even though its response away from that state remains anisotropic. A single scalar correction affine in the determinant can then cancel the complete reference stress without tying that correction to the independently adjustable volumetric stiffness. The feature family thus gains flexibility while retaining an analytically controlled reference state.

The family is coupled to a smooth ICNN and an integrated-hat spline ICKAN, both convex and componentwise nondecreasing in the features. For either core, the proof covers the complete energy, including normalization and growth terms, and establishes objectivity, an exactly stress-free reference, two-dimensional polyconvexity, and the specified growth limits. The spline realization uses analytically matched tails. Global nonnegative energy is addressed separately by a sufficient lower bound evaluated for the trained parameters. The contribution builds on established convex-network and affine-determinant normalization principles; it is their realization with the paired anisotropic family and its accompanying analytical conditions.

To determine what this construction offers in practice, we compare it with direct stress regression, an unrestricted energy network, and the reduced evaluators developed from the same RVE. These alternatives separate the consequences of imposing a potential, restricting its curvature, and retaining microscopic state information. POD, latent-space closure, ECM, and manifold-adaptive cubature are established ingredients of the comparison rather than new reduction algorithms proposed here. Figure 1 summarizes the models and their subsequent structural deployment.

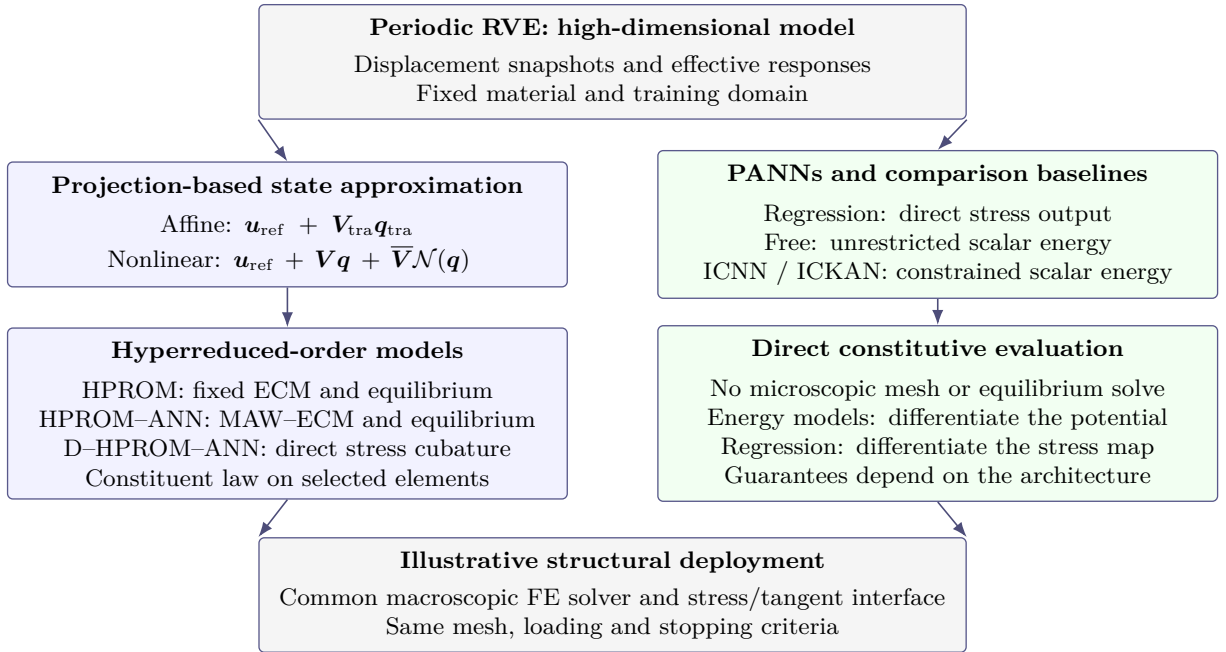


Figure 1: Structural deployment of the PANN and PROM-based surrogates developed or assembled for the RVE study. PROM-based models reconstruct microscopic displacements and evaluate the constituent law; energy-based PANNs directly represent the homogenized potential. Regression and Free provide comparison baselines. The diagram distinguishes reduced equilibrium solves from direct evaluation.

1.5 Assessment strategy and scope

The examples address two successive questions. First, can the constrained energy represent the anisotropic response of the periodic cell, and do its mechanical properties distinguish it from an accurate but less constrained fit? The constitutive assessment therefore combines held-out response errors with tests of work around closed strain cycles and sampled rank-one curvature, following the broader motivation to examine constitutively relevant derivatives as well as fitting errors [42]. These numerical checks are kept separate from the construction proof and the saved-parameter nonnegative-energy bound. In particular, sampled positive curvatures do not establish polyconvexity.

Second, how do those constitutive approximations behave within a nonlinear structural calculation? A tensile coupon supplies this deployment test. In the first-order two-level finite element formulation, denoted FE^2 , each macroscopic integration point queries the same RVE response. Replacing that query by each surrogate propagates its approximation through a common equilibrium problem. A nested full-order

calculation (FOM–FE²) supplies the reference, and field-level comparisons complement aggregate displacement and stress errors [43]. The example thus connects material-level assessment to structural accuracy and online cost without making the development of a new FE² scheme the subject of the paper.

The interpretation follows the purpose of each test. Nonzero closed-cycle work identifies a lack of integrability in Regression and the three deployed HPROM variants, even when their structural solutions converge. The extrapolative Free-model curvature witness, in contrast, lacks a converged FOM reference and is reported as a property of that learned model, not as an FOM-confirmed artificial instability. The comparison is therefore not organized around making every unconstrained model fail, but around identifying which statements are supported by its construction and which require independent evidence.

The present evidence concerns one periodic material under plane-strain kinematics, one prescribed training domain, and one structural loading path. The coupon uses the same RVE, so it tests deployment rather than providing a second material validation. FOM–FE² agreement measures surrogate error within the adopted homogenization model, not its accuracy against a structure resolving every microscopic cell. The study assesses the feature family, normalization, and core together; isolating their individual contributions requires matched-budget ablations. Likewise, the reported online times compare the implemented workflows at the stated process/thread counts and repetition levels, excluding offline data generation and training. They do not quantify memory savings or an offline-inclusive break-even point. Within this scope, the aim is to establish the usefulness and constitutive properties of the proposed representation, not universality across microstructures, general three-dimensional completeness, or guaranteed robustness under microscopic instability.

Section 2 defines the homogenization problem and work-conjugate variables. Section 3 develops the PROM-based surrogates. Section 4 introduces the admissibility criteria, constructs the learned energies, and proves their properties before specifying the implementation checks. Sections 5 and 6 report the constitutive and structural assessments, respectively; Section 7 closes the paper.

2 Finite-strain homogenization and the reference problem

The surrogates considered here approximate the same constitutive problem: given a macroscopic deformation, determine the effective response of a periodic RVE. The full-order model obtains this response by solving microscopic equilibrium. We first define its input, boundary conditions, and effective outputs, and then identify the stress–tangent interface needed for structural deployment. The cell geometry and material parameters are specified in the numerical examples.

2.1 Kinematics and work-conjugate variables

Let $\bar{\mathbf{F}} \in \text{GL}^+(2)$ denote the prescribed in-plane macroscopic deformation gradient. Its associated right Cauchy–Green tensor and Green–Lagrange strain are

$$\mathbf{C} = \bar{\mathbf{F}}^T \bar{\mathbf{F}}, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad J = \det \bar{\mathbf{F}} > 0. \quad (1)$$

Here $\mathbf{C}$ and $\mathbf{E}$ describe the macroscopic state; the subscript μ will distinguish microscopic quantities. We adopt plane strain: the corresponding three-dimensional gradient is $\text{diag}(\bar{\mathbf{F}}, 1)$, and the same embedding applies locally to $\mathbf{F}_\mu$. This constrains the out-of-plane deformation, not the out-of-plane stress.

The constitutive input is the three-component engineering strain vector. With $\bar{\mathbf{S}}$ the effective second Piola stress, the work-conjugate convention is

$$\mathbf{e} = (E_{11}, E_{22}, 2E_{12})^T, \quad \mathbf{s} = (\bar{S}_{11}, \bar{S}_{22}, \bar{S}_{12})^T, \quad \bar{\mathbf{S}} : d\mathbf{E} = \mathbf{s} \cdot d\mathbf{e}. \quad (2)$$

Thus $\gamma_{12} = 2E_{12}$ is paired with $\bar{S}_{12}$, with no additional factor of two in the stress vector. For a differentiable effective energy density $\bar{W}(\mathbf{e})$, this convention gives $\mathbf{s} = \partial_{\mathbf{e}} \bar{W}$ and $\mathbf{D} = \partial_{\mathbf{e}} \mathbf{s}$. It is used consistently in training, work integrals, and tangent checks.

An admissible strain input must satisfy $\mathbf{I} + 2\mathbf{E} \succ 0$. We query the RVE with the representative

$$\bar{\mathbf{F}}(\mathbf{e}) = \sqrt{\mathbf{I} + 2\mathbf{E}}, \quad (3)$$

where the square root is symmetric positive definite. Objectivity allows a superposed macroscopic rotation to be removed without changing the energy or second Piola stress. This choice does not assume isotropy: material directions remain fixed in the reference configuration. Nor does it require the local microscopic gradients to be symmetric.

2.2 Periodic microscopic equilibrium

Let Y denote the reference periodic cell and $Y_s \subset Y$ its solid part. The prescribed $\bar{\mathbf{F}}$ determines the affine part of the microscopic deformation; the fluctuation $\mathbf{w}$ is unknown:

$$\boldsymbol{\varphi}(\mathbf{X}) = \bar{\mathbf{F}}\mathbf{X} + \mathbf{w}(\mathbf{X}), \quad \mathbf{F}_\mu = \bar{\mathbf{F}} + \nabla \mathbf{w}, \quad (4)$$

with gradients taken in the reference configuration. Periodicity requires $\mathbf{w}(\mathbf{X}^+) = \mathbf{w}(\mathbf{X}^-)$ at paired points on opposite cell faces. Consequently, the microscopic displacement $\mathbf{u}_\mu = \boldsymbol{\varphi} - \mathbf{X}$ satisfies

$$\mathbf{u}_\mu(\mathbf{X}^+) - \mathbf{u}_\mu(\mathbf{X}^-) = (\bar{\mathbf{F}} - \mathbf{I})(\mathbf{X}^+ - \mathbf{X}^-). \quad (5)$$

The finite-strain displacement jump therefore depends on $\bar{\mathbf{F}} - \mathbf{I}$, not directly on $\mathbf{E}$. Setting one fluctuation value to zero removes the arbitrary translation; it does not clamp the entire cell boundary. The internal void surface is traction-free.

Let $\psi_\mu(\mathbf{X}, \mathbf{F}_\mu)$ be the solid's objective hyperelastic energy per unit reference volume and $\mathbf{P}_\mu = \partial_{\mathbf{F}_\mu} \psi_\mu$ its first Piola stress. In the absence of microscopic body forces, equilibrium requires

$$\int_{Y_s} \mathbf{P}_\mu : \nabla \delta \mathbf{w} \, dV = 0 \quad (6)$$

for every admissible periodic test fluctuation. Periodic kinematics and the associated traction balance provide the standard work-consistent cell problem [1]. After finite element discretization and elimination of the periodic constraints, let $\mathbf{a}$ collect the independent fluctuation coordinates. The full-order equilibrium problem is

$$\mathbf{r}(\mathbf{a}, \mathbf{e}) = \partial_{\mathbf{a}} \Pi_\mu(\mathbf{a}, \mathbf{e}) = \mathbf{0}, \quad \Pi_\mu = \int_{Y_s} \psi_\mu(\mathbf{X}, \mathbf{F}_\mu) \, dV, \quad (7)$$

where the strain dependence enters through Eq. (3). Solving Eq. (7) gives $\mathbf{a}^*(\mathbf{e})$ and hence the microscopic fields. The microscopic deformation must also remain orientation-preserving, $\det \mathbf{F}_\mu > 0$.

2.3 Effective energy and stress

On the selected equilibrium branch, the effective energy and stresses are

$$\bar{W}(\mathbf{e}) = \frac{\Pi_\mu(\mathbf{a}^*(\mathbf{e}), \mathbf{e})}{|Y|}, \quad \bar{\mathbf{P}} = \frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu \, dV, \quad \bar{\mathbf{S}} = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}. \quad (8)$$

The normalizing volume $|Y|$ includes the void: these are properties of the porous cell, not of its solid fraction alone. For the two-dimensional discretization, the same reference thickness enters the integration weights and the cell volume, and cancels in their ratio. The stress conversion in Eq. (8) is performed *after* averaging the first Piola stress. In general, averaging microscopic second Piola stresses does not produce $\bar{\mathbf{S}}$, because $\mathbf{F}_\mu$ varies within the cell.

The connection between energy and averaged stress follows from equilibrium. For a macroscopic variation, $\delta \mathbf{F}_\mu = \delta \bar{\mathbf{F}} + \nabla \delta \mathbf{w}$; Eq. (6) removes the fluctuation contribution and yields

$$\frac{1}{|Y|} \int_{Y_s} \mathbf{P}_\mu : \delta \mathbf{F}_\mu \, dV = \bar{\mathbf{P}} : \delta \bar{\mathbf{F}} = \bar{\mathbf{S}} : \delta \mathbf{E} = \mathbf{s} : \delta \mathbf{e}. \quad (9)$$

This is the Hill–Mandel work-consistency condition [1]; it also establishes $\mathbf{s} = \partial_{\mathbf{e}} \bar{W}$ on a smooth branch. The subsequent constitutive constructions omit overbars when the effective, rather than microscopic, response is unambiguous.

The Newton solver follows an equilibrium branch, possibly through load continuation. Convergence establishes stationarity, not global minimization of Π_μ or uniqueness of the cell response. We use the resulting smooth branch as the reference constitutive response over the investigated states; we do not assume that microscopic polyconvexity guarantees a polyconvex homogenized energy.

2.4 Consistent constitutive tangent

The tangent must account for the change in microscopic equilibrium as the input strain changes. To make this dependence explicit, write the stress evaluator before substitution of the equilibrium state as $\hat{\mathbf{s}}(\mathbf{a}, \mathbf{e})$, so that $\mathbf{s}(\mathbf{e}) = \hat{\mathbf{s}}(\mathbf{a}^*(\mathbf{e}), \mathbf{e})$. If $\mathbf{r}_{,\mathbf{a}}$ is nonsingular on the selected branch, implicit differentiation of Eq. (7) gives

$$\frac{d\mathbf{a}^*}{d\mathbf{e}} = -\mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}, \quad \mathbf{D} = \hat{\mathbf{s}}_{,\mathbf{e}} - \hat{\mathbf{s}}_{,\mathbf{a}} \mathbf{r}_{,\mathbf{a}}^{-1} \mathbf{r}_{,\mathbf{e}}. \quad (10)$$

All partial derivatives are evaluated at equilibrium, holding the other argument fixed. The first term differentiates the stress at fixed fluctuation coordinates; the second accounts for microscopic relaxation. Omitting that correction generally does not give the derivative of the deployed constitutive response. For the smooth hyperelastic branch, the result is also $\partial_{ee}^2 \bar{W}$.

No explicit matrix inverse is needed in Eq. (10): the sensitivities follow from linear systems with the converged microscopic tangent. The reference implementation uses one nonlinear equilibrium solution followed by three sensitivity right-hand sides, one per strain component. Independent finite-difference checks perturb the input and re-solve equilibrium. Such checks assess the constitutive linearization, not the physical accuracy of the homogenization assumptions. The same consistency requirement applies to each surrogate: its tangent must differentiate its own stress evaluation.

2.5 Macroscopic equilibrium and the constitutive interface

Structural deployment provides an application of this constitutive map. Let $\mathbf{U}$ collect the free macroscopic nodal displacements, and let $\mathbf{e}_g(\mathbf{U})$ be the engineering Green strain at integration point g . With $\mathbf{B}_g = \partial \mathbf{e}_g / \partial \mathbf{U}$, reference-volume integration weights Ω_g , and dead external loads $\mathbf{f}_{\text{ext}}$, equilibrium reads

$$\mathbf{R}_M(\mathbf{U}) = \sum_g \Omega_g \mathbf{B}_g^T \mathbf{s}_g(\mathbf{e}_g(\mathbf{U})) - \mathbf{f}_{\text{ext}} = \mathbf{0}. \quad (11)$$

Its tangent contains material and geometric contributions,

$$\mathbf{K}_M = \sum_g \Omega_g \left[\mathbf{B}_g^T \mathbf{D}_g \mathbf{B}_g + \sum_{\alpha=1}^3 (s_g)_\alpha \frac{\partial^2 (e_g)_\alpha}{\partial \mathbf{U} \partial \mathbf{U}} \right]. \quad (12)$$

The first term uses the constitutive tangent; the second arises because the Green strain is nonlinear in the displacements. A structural Newton solver therefore needs both $\mathbf{s}_g$ and $\mathbf{D}_g$, irrespective of how the constitutive response is computed.

In FOM-FE², each constitutive call invokes the full periodic RVE problem. The iterative HPROM variants replace it with projected equilibrium on reduced coordinates and hyperreduced integration. The direct decoder and learned constitutive models bypass microscopic equilibrium iterations. These alternatives are developed in the following sections; all supply the same stress–tangent interface while the macroscopic mesh, loading, and equilibrium algorithm remain unchanged.

The nested FOM-FE² solution is the structural reference for measuring surrogate errors at fixed macro and micro discretizations. It is not a direct numerical simulation resolving every pore in the coupon. Homogenization error, boundary-layer effects, and mesh-discretization error are therefore distinct from the surrogate errors reported below. The cell itself is an ideal periodic material model, without a claim of statistical representativeness for a random microstructure. The suffix FE² in archived model names identifies the common structural test; a direct neural constitutive evaluation is not a nested microscopic finite element solve.

3 Projection-based reduced-order models

The intrusive route reduces the cost of the RVE problem while retaining evaluations of the microscopic constitutive law. Two approximations play different roles. A reduced displacement representation decreases the number of equilibrium unknowns; hyperreduction decreases the number of element contributions evaluated to assemble the reduced equations and effective outputs. We first construct affine and nonlinear-manifold PROMs, then introduce fixed and manifold-adaptive cubature. Finally, we distinguish solving reduced equilibrium from evaluating the same decoder directly.

3.1 The constrained high-dimensional model

We follow the nomenclature of Ref. [18]: the high-dimensional model (HDM) has state dimension N ; a PROM solves for generalized coordinates in a reduced approximation space; an HPROM additionally hyperreduces the evaluation of its projected operators. HDM and FOM denote the same unreduced microscopic discretization here; FOM is retained in the archived name FOM-FE².

It is useful to distinguish the physical nodal displacement vector $\mathbf{d} \in \mathbb{R}^{N_{\text{dof}}}$ from the independent state $\mathbf{u} \in \mathbb{R}^N$ obtained after enforcing periodicity and removing rigid translation. In the deployed convention,

$$\mathbf{d}(\mathbf{u}, \mathbf{e}) = \mathbf{T} \mathbf{u} + \mathbf{g}_{\text{per}}(\mathbf{e}), \quad \mathbf{R}(\mathbf{u}; \mathbf{e}) = \mathbf{T}^T \mathbf{f}_{\text{int}}(\mathbf{d}(\mathbf{u}, \mathbf{e})) = \mathbf{0}. \quad (13)$$

Here $\mathbf{T}$ identifies periodic degrees of freedom, and the lifting $\mathbf{g}_{\text{per}}$ supplies the strain-dependent displacement jumps and anchor values defined in Section 2. Thus any independent state $\mathbf{u}$ produces a physical field satisfying the prescribed kinematic constraints. If $\mathbf{K}_{\text{phys}} = \partial_{\mathbf{a}} \mathbf{f}_{\text{int}}$ is the assembled physical stiffness, the constrained HDM tangent is $\mathbf{K} = \partial_{\mathbf{u}} \mathbf{R} = \mathbf{T}^T \mathbf{K}_{\text{phys}} \mathbf{T}$. We use the internal-force sign convention for all residuals.

The snapshots used below retain the affine part of the independent displacement. They therefore differ from the fluctuation coordinates $\mathbf{a}$ of Eq. (7): if $\mathbf{u}_{\text{aff}}(\mathbf{e})$ is the affine displacement at the independent nodes, then $\mathbf{a} = \mathbf{u} - \mathbf{u}_{\text{aff}}(\mathbf{e})$. This is a change of coordinates, not a change of boundary conditions. The snapshot convention and lifting must be used together; throughout this section, $\mathbf{u}$ denotes the affine-retaining state of Eq. (13).

We retain this part to preserve load-correlated coordinates for the strain-informed identification below; preliminary modal-coordinate fits favored this convention over fluctuation-only snapshots. This is a representation choice, not a requirement of periodicity or of direct evaluation: a fluctuation decoder could instead be combined with an explicit affine lifting. Although $\mathbf{u}_{\text{aff}} = (\bar{\mathbf{F}}(\mathbf{e}) - \mathbf{I})\mathbf{X}$ is nonlinear in $\mathbf{e}$, the symmetric representative $\bar{\mathbf{F}}$ makes these affine fields linear combinations of at most three fixed nodal fields. The convention can nevertheless change the snapshot spectrum and its normalization. The retained POD dimension therefore characterizes the chosen representation and tolerance, not an intrinsic material dimension.

3.2 POD approximation and the conventional PROM

Let $\mathbf{u}^s$ be the HDM solution at a sampled macroscopic strain $\mathbf{e}^s$, with $s = 1, \dots, N_s$. Given a reference state $\mathbf{u}_{\text{ref}}$, form

$$\mathbf{S}_{\text{snap}} = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}}, \dots, \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}] = \mathbf{U}_S \boldsymbol{\Sigma}_S \mathbf{Y}_S^T. \quad (14)$$

The subscript distinguishes the snapshot matrix from the second Piola stress $\mathbf{S}$. Proper orthogonal decomposition (POD) selects the columns of $\mathbf{U}_S$ associated with the largest singular values to form a reduced-order basis (ROB). For an orthonormal n_{tra} -column basis $\mathbf{V}_{\text{tra}}$, the conventional affine approximation is

$$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}, \quad \mathbf{q}_{\text{tra}} \in \mathbb{R}^{n_{\text{tra}}}, \quad \mathbf{V}_{\text{tra}}^T \mathbf{V}_{\text{tra}} = \mathbf{I}. \quad (15)$$

Instead of adjusting all N displacement coordinates independently, the model adjusts $n_{\text{tra}} \ll N$ modal amplitudes. The term *linear PROM* refers to this affine approximation space; the constitutive law and reduced equilibrium equations remain nonlinear.

For a Euclidean POD, the relative squared snapshot reconstruction error is

$$\epsilon_{\text{POD}}^2(n_{\text{tra}}) = \frac{\sum_{j > n_{\text{tra}}} \sigma_j^2}{\sum_j \sigma_j^2}. \quad (16)$$

This is a compression criterion for the supplied snapshots, not an error bound for stresses at unsampled strains. Force evaluation, constitutive differentiation, and equilibrium conditioning can make small displacement components important.

At a fixed input $\mathbf{e}$, admissible reduced variations are $\delta \tilde{\mathbf{u}} = \mathbf{V}_{\text{tra}} \delta \mathbf{q}_{\text{tra}}$. Requiring the HDM residual to perform zero virtual work against these variations gives the Galerkin equations and their tangent:

$$\mathbf{r}_{\text{tra}}(\mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{V}_{\text{tra}}^T \mathbf{R}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}; \mathbf{e}) = \mathbf{0}, \quad \mathbf{K}_{\text{tra}} = \mathbf{V}_{\text{tra}}^T \mathbf{K} \mathbf{V}_{\text{tra}}. \quad (17)$$

The system has n_{tra} unknowns, but assembling $\mathbf{R}$ and $\mathbf{K}$ on every microscopic element still scales with the full discretization. This is the cost removed by hyperreduction, not by POD alone.

3.3 Nonlinear reduction through latent-space closure

For a chosen displacement representation, a low-dimensional loading family may occupy a much larger linear approximation space. Latent-space closure exploits this distinction without attributing the required mode count solely to material physics: it retains a sufficiently rich displacement span, but learns dependencies among its modal amplitudes so that fewer of them remain equilibrium unknowns [17, 18]. Let $n_{\text{tot}} = n + \bar{n}$ be the dimension of that span, partitioned into mutually orthogonal primary and secondary ROB:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad \bar{\mathbf{V}} \in \mathbb{R}^{N \times \bar{n}}, \quad (18)$$

$$\mathbf{V}^T \bar{\mathbf{V}} = \mathbf{0}, \quad \mathbf{V}^T \mathbf{V} = \mathbf{I}, \quad \bar{\mathbf{V}}^T \bar{\mathbf{V}} = \mathbf{I}. \quad (19)$$

The associated primary and secondary generalized coordinates are

$$\mathbf{q}^s = \mathbf{V}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{q}}^s = \bar{\mathbf{V}}^T (\mathbf{u}^s - \mathbf{u}_{\text{ref}}). \quad (20)$$

The primary coordinates $\mathbf{q}$ are retained as independent unknowns. Instead of also solving for $\bar{\mathbf{q}}$, we learn the secondary-coordinate closure $\mathcal{N}: \mathbb{R}^n \rightarrow \mathbb{R}^{\bar{n}}$ from pairs $(\mathbf{q}^s, \bar{\mathbf{q}}^s)$. This closure can be constructed using ANNs, Gaussian process regression (GPR), or radial basis function (RBF) interpolation [18]. We use an ANN to maintain a common neural parametrization across the learned components of this study, not because latent-space closure requires one. With network parameters $\boldsymbol{\eta}$, the resulting decoder and its tangent are

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}(\mathbf{q}; \boldsymbol{\eta}), \quad \mathbf{B}(\mathbf{q}) = \partial_{\mathbf{q}}\tilde{\mathbf{u}} = \mathbf{V} + \bar{\mathbf{V}}\mathcal{N}_{,\mathbf{q}}. \quad (21)$$

The correction $\bar{\mathbf{V}}\mathcal{N}(\mathbf{q}; \boldsymbol{\eta})$ turns the affine primary approximation into a nonlinear manifold. The ANN predicts neither stresses nor a macroscopic energy: it predicts the $\bar{n}$ secondary amplitudes used to reconstruct a microscopic displacement. These modes are discarded only as independent equilibrium unknowns, not from the reconstructed field. This is the PROM-ANN construction; no hyperreduction has yet been introduced.

For orthonormal blocks, the error at a snapshot separates into closure error and a component outside the total ROB:

$$\|\mathbf{u}^s - \tilde{\mathbf{u}}(\mathbf{q}^s)\|_2^2 = \|\bar{\mathbf{q}}^s - \mathcal{N}(\mathbf{q}^s)\|_2^2 + \|(\mathbf{I} - \mathbf{V}_{\text{tot}}\mathbf{V}_{\text{tot}}^T)(\mathbf{u}^s - \mathbf{u}_{\text{ref}})\|_2^2. \quad (22)$$

This explains the low-dimensional targets: the ANN approximates secondary coordinates, while the ROB performs the displacement reconstruction. It also separates limitations that cannot be repaired by the same operation. Increasing network capacity cannot recover a component excluded from $\mathbf{V}_{\text{tot}}$; adding secondary modes cannot make a non-single-valued primary-to-secondary relation single-valued.

The closure is fitted by minimizing the unscaled secondary-coordinate mean-square error,

$$\mathcal{L}_{\text{cl}}(\boldsymbol{\eta}) = \frac{1}{|\mathcal{I}_{\text{fit}}|\bar{n}} \sum_{s \in \mathcal{I}_{\text{fit}}} \|\mathcal{N}(\mathbf{q}^s; \boldsymbol{\eta}) - \bar{\mathbf{q}}^s\|_2^2. \quad (23)$$

Input standardization is absorbed into the closure map, but the secondary outputs are not standardized mode by mode. Because $\bar{\mathbf{V}}$ is orthonormal, this loss measures the independent-displacement error within the secondary space, up to its constant normalization. It is not a strain-energy norm or a stress-error guarantee. The implemented network uses smooth tanh activations, allowing the decoder derivatives required by projection and constitutive differentiation.

For the comparison below, the affine and nonlinear models share the same retained span, with $n_{\text{tot}} = n_{\text{tra}}$. The nonlinear model represents an n -dimensional subset of this space, not every displacement the affine PROM can represent. The dimensions and reconstruction diagnostics are reported with the RVE example.

3.3.1 Strain-informed primary coordinates

A useful primary coordinate system must distinguish the displacement states whose secondary coordinates are to be predicted. The leading POD coefficients maximize captured snapshot energy, but that criterion alone does not ensure a well-conditioned primary-to-secondary relation. We use the input-informed decoder construction introduced alongside MAW-ECM in Ref. [32], with the three macroscopic strain components as inputs and $n = 3$. This construction is independent of the cubature method: it rotates the retained POD span to identify coordinates approximately aligned with those inputs.

Let $\boldsymbol{\Phi} \in \mathbb{R}^{N \times n_{\text{tot}}}$ be the initial orthonormal POD basis, and collect its snapshot coefficients and the corresponding strain inputs as

$$\mathbf{Q}_{\text{POD}} = \boldsymbol{\Phi}^T \mathbf{S}_{\text{snap}}, \quad \mathbf{M} = [\mathbf{e}^1, \dots, \mathbf{e}^{N_s}]. \quad (24)$$

The least-squares identification

$$\mathbf{T}_m \in \arg \min_{\mathbf{H}} \|\mathbf{H}\mathbf{Q}_{\text{POD}} - \mathbf{M}\|_F^2, \quad \mathbf{T}_m = \mathbf{M}\mathbf{Q}_{\text{POD}}^\dagger, \quad (25)$$

uses the Moore–Penrose pseudoinverse to select the minimum-norm solution. A compact SVD,

$$\boldsymbol{\Phi}\mathbf{T}_m^T = \mathbf{V}\boldsymbol{\Sigma}_m\mathbf{Z}_m^T, \quad \mathbf{A}_m = \boldsymbol{\Sigma}_m^{-1}\mathbf{Z}_m^T, \quad (26)$$

defines the primary orthonormal block, assuming rank three. The secondary block is an orthonormal complement *within* the retained POD span, ordered by its snapshot singular values. The coordinates used in the implementation are $\boldsymbol{\xi} = \mathbf{A}_m^{-1}\mathbf{q}$, with closure $\hat{\mathcal{N}}(\boldsymbol{\xi})$. Consequently, Eq. (21) becomes

$$\tilde{\mathbf{u}}(\boldsymbol{\xi}) = \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{A}_m\boldsymbol{\xi} + \bar{\mathbf{V}}\hat{\mathcal{N}}(\boldsymbol{\xi}), \quad \mathcal{N}(\mathbf{q}) = \hat{\mathcal{N}}(\mathbf{A}_m^{-1}\mathbf{q}). \quad (27)$$

The SVD gives $\mathbf{A}_m^{-1} = \mathbf{Z}_m\boldsymbol{\Sigma}_m$, hence $\boldsymbol{\xi}^s = \mathbf{T}_m\mathbf{Q}_{\text{POD}}^s$ for the projected snapshots. Their proximity to $\mathbf{e}^s$ is a fitted relation, not a kinematic identity. In particular, the input $\mathbf{e}$ prescribes the periodic lifting, while $\boldsymbol{\xi}$ parametrizes the unknown displacement. The iterative model solves for $\boldsymbol{\xi}$ at fixed $\mathbf{e}$; it does not impose their equality. In these coordinates its decoder tangent is $\mathbf{B}_{\boldsymbol{\xi}} = \mathbf{B}\mathbf{A}_m$, so the change of variables preserves the projected equilibrium problem. Below, formulas written in $\mathbf{q}$ are evaluated with this transformation when using $\boldsymbol{\xi}$.

3.3.2 Equilibrium on the nonlinear manifold

Learning the decoder specifies where the reduced solution may lie, but does not determine its location for a new input. For an iterative PROM-ANN, that location is selected by projected equilibrium. At fixed $\mathbf{e}$, the virtual displacement is $\delta \tilde{\mathbf{u}} = \mathbf{B}(\mathbf{q})\delta \mathbf{q}$, giving

$$\mathbf{r}(\mathbf{q}; \mathbf{e}) = \mathbf{B}(\mathbf{q})^T \mathbf{R}(\tilde{\mathbf{u}}(\mathbf{q}); \mathbf{e}) = \mathbf{0}. \quad (28)$$

Its exact derivative is

$$\frac{\partial \mathbf{r}}{\partial \mathbf{q}} = \mathbf{B}^T \mathbf{K} \mathbf{B} + \sum_{j=1}^N R_j \frac{\partial^2 \tilde{u}_j}{\partial \mathbf{q} \partial \mathbf{q}}. \quad (29)$$

The second term accounts for manifold curvature. It need not vanish at reduced equilibrium, because $\mathbf{B}^T \mathbf{R} = \mathbf{0}$ does not imply $\mathbf{R} = \mathbf{0}$. Dropping it defines an approximate iteration matrix, not the exact Jacobian.

The latent closure does not prescribe the projection method. Least-squares Petrov-Galerkin (LSPG) instead minimizes a discrete residual norm and uses a test space involving the residual Jacobian and decoder tangent [18, 20, 43]. We use Galerkin virtual work for the present periodic hyperelastic problem. This choice retains the primary/secondary construction without identifying Galerkin and LSPG as the same reduced equations.

3.4 Hyperreduction of the projected equations

Although the PROMs reduce the number of equilibrium unknowns, evaluating their projected equations still requires full-mesh contributions. ECM addresses this remaining cost by replacing the element sum with a sparse weighted sum [29]. This is a second approximation, applied after choosing the displacement representation.

3.4.1 Fixed elementwise cubature

Let $\mathcal{E}$ be the microscopic element set and $\mathbf{L}_e$ extract element nodal values from $\mathbf{d}$. At fixed macroscopic input, define the reconstructed element state and its reduced tangent by

$$\mathbf{d}_e = \mathbf{L}_e(\mathbf{T}\tilde{\mathbf{u}} + \mathbf{g}_{\text{per}}), \quad \mathbf{B}_e = \mathbf{L}_e \mathbf{T} \mathbf{B}. \quad (30)$$

The full reduced residual is $\sum_{e \in \mathcal{E}} \mathbf{h}_e$, where $\mathbf{h}_e = \mathbf{B}_e^T \mathbf{f}_{\text{int},e}$. Fixed ECM approximates it as

$$\mathbf{r}_{\text{ECM}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e \mathbf{h}_e(\mathbf{q}; \mathbf{e}) = \mathbf{0}, \quad \mathcal{Z}_r \subset \mathcal{E}, \quad w_e \geq 0. \quad (31)$$

For the conventional HPROM, $\mathbf{B}$ is replaced by $\mathbf{V}_{\text{tra}}$ and the unknown is $\mathbf{q}_{\text{tra}}$. There is no ANN in this tier. A selected element retains its ordinary within-element quadrature. Consequently, selected elements, selected Gauss points, and independent coordinates are distinct counts.

To build the rule offline, we must retain the individual element contributions before summing them. Let $N_{\text{el}} = |\mathcal{E}|$, and let $\mathbf{c}_e^s \in \mathbb{R}^d$ collect the d integrand components of element e at training state s , for $s = 1, \dots, N_c$. For residual cubature, $\mathbf{c}_e^s = \mathbf{h}_e^s$, with $d = n_{\text{tra}}$ for the affine model and $d = n$ for the nonlinear manifold. Output cubature instead uses the four components of the element contribution to the average first Piola tensor:

$$\mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu,g}^s, \quad \mathbf{c}_e^s = \text{vec } \mathbf{p}_e^s = [p_{e,11}^s \quad p_{e,12}^s \quad p_{e,21}^s \quad p_{e,22}^s]^T, \quad d = 4, \quad (32)$$

where ω_g is the reference quadrature measure, $\mathbf{P}_{\mu,g}^s$ is the microscopic first Piola stress, and vec uses the order (11, 12, 21, 22). Residual and output rules are built separately. Physical integration measures are already included in $\mathbf{c}_e^s$; the unknown cubature weights are element multipliers.

The raw integrand matrix arranges one element per row and one component of one training state per column:

$$\mathbf{M}_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \cdots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \cdots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \in \mathbb{R}^{N_{\text{el}} \times (dN_c)}, \quad (\mathbf{M}_{\text{int}})_{e,(s-1)d+\alpha} = c_{e,\alpha}^s. \quad (33)$$

Thus the contribution vectors are concatenated across states, not assembled across elements. Compressing this matrix with a truncated SVD gives $\mathbf{M}_{\text{int}} \simeq \mathbf{U}_{\text{cub}} \mathbf{\Sigma}_{\text{cub}} \mathbf{V}_{\text{cub}}^T$, where $\mathbf{U}_{\text{cub}} \in \mathbb{R}^{N_{\text{el}} \times r_c}$ contains the r_c retained integration modes and $\mathbf{U}_{\text{cub}}^T \mathbf{U}_{\text{cub}} = \mathbf{I}$. These are modes of the *integrands*, not displacement POD modes.

Transposing $\mathbf{U}_{\text{cub}}$ makes each candidate element a column of the fitting system. The deployed ECM implementation adds the normalized part of the constant vector not already represented by that basis:

$$\mathbf{p}_{\perp} = (\mathbf{I} - \mathbf{U}_{\text{cub}} \mathbf{U}_{\text{cub}}^T) \mathbf{1}, \quad \mathbf{c}_{\perp} = \frac{\mathbf{p}_{\perp}}{\|\mathbf{p}_{\perp}\|_2}, \quad \mathbf{G}_{\text{cub}} = \begin{bmatrix} \mathbf{U}_{\text{cub}}^T \\ \mathbf{c}_{\perp}^T \end{bmatrix} \in \mathbb{R}^{(r_c+1) \times N_{\text{el}}}. \quad (34)$$

This expression assumes $\|\mathbf{p}_{\perp}\|_2 > 0$, as in the deployed rules; if the constant is already in the retained span, the additional constraint is redundant. Exact reproduction of these rows implies $\mathbf{1}^T \mathbf{w} = N_{\text{el}}$. Orthogonalizing the constant row avoids duplicating its component in the retained integration space. Figure 2(a) shows the passage from sampled contributions to this compressed system. The fitting problem on a selected support has the form

$$\min_{\mathbf{w} \geq 0} \|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2, \quad \mathbf{b}_{\text{cub}} = \mathbf{G}_{\text{cub}} \mathbf{1}, \quad (35)$$

Here unselected weights are zero. Nonnegative least squares alone does not select a sparse rule: $\mathbf{w} = \mathbf{1}$ is the full-mesh solution. ECM combines fitting with greedy support selection until the relative integration residual satisfies

$$\frac{\|\mathbf{G}_{\text{cub}} \mathbf{w} - \mathbf{b}_{\text{cub}}\|_2}{\|\mathbf{b}_{\text{cub}}\|_2} \leq \varepsilon_{\text{ECM}}. \quad (36)$$

The tolerance ε_{ECM} applies to the compressed integration constraints. It is distinct from the integrand-SVD truncation tolerance ε_{SVD} , which determines how many constraints are retained, and from the subsequent error in an effective output. Training retains the individual contributions even when their assembled equilibrium sum vanishes. Further details of integration-basis compression and greedy selection are given in Ref. [29].

The deployed element-multiplier rules also impose $\sum_e w_e = |\mathcal{E}|$, with physical quadrature measures already contained in the element integrands. This is an element-count normalization, not by itself exact physical-volume integration on an unequal-volume mesh. Once constructed, both support and weights remain fixed online.

3.4.2 Manifold-adaptive weights

A single weight vector must satisfy integration constraints collected from every training state. Allowing the weights to depend on the reduced state relaxes that shared-weight requirement while retaining a common sparse support. This is the principle of MAW-ECM [32]. We use it with elementwise sampling and neural weight fields; the state representation remains the PROM-ANN decoder. The adaptive residual is

$$\mathbf{r}_{\text{MAW}}(\mathbf{q}; \mathbf{e}) = \sum_{e \in \mathcal{Z}_r} w_e(\mathbf{q}) \mathbf{B}_e(\mathbf{q})^T \mathbf{f}_{\text{int},e}(\mathbf{d}_e) = \mathbf{0}. \quad (37)$$

Fixed ECM first supplies a candidate support. Adaptive-weight pruning then removes elements while readjusting state-specific weights subject to the integration constraints and nonnegativity, until a prescribed target support size is reached. The neural fields are fitted on this pruned support, not on a fixed-ECM support selected directly at the final size. The pruning formulation, including its local and graph-coupled regularization mechanisms, is detailed in Ref. [32]; the available support-generation script activates the local positivity-enforcing phase from the first elimination. Its settings and the distinction between support selection and field fitting are recorded with the numerical example.

Pruning determines *which elements remain*; it does not yet provide a neural map for evaluating their weights at a new state. After pruning, we therefore train an additional ANN whose outputs determine the cubature weights. This is a separate network from the displacement closure $\mathcal{N}$, which has already been trained and remains fixed during weight fitting. Because equilibrium and stress use separate integration rules, the deployed model actually contains two such weight ANNs, denoted $\mathcal{N}_r$ and $\mathcal{N}_s$. Their roles are

$$\begin{aligned} \bar{\mathbf{q}}(\mathbf{q}) &= \mathcal{N}(\mathbf{q}; \boldsymbol{\eta}) \in \mathbb{R}^{\bar{n}}, & \text{secondary-coordinate ANN,} \\ \boldsymbol{\ell}_r(\mathbf{q}) &= \mathcal{N}_r(\mathbf{q}; \boldsymbol{\theta}_r) \in \mathbb{R}^{m_r}, & \text{residual-weight ANN,} \\ \boldsymbol{\ell}_s(\mathbf{q}) &= \mathcal{N}_s(\mathbf{q}; \boldsymbol{\theta}_s) \in \mathbb{R}^{m_s}, & \text{stress-weight ANN,} \end{aligned} \quad (38)$$

where $m_r = |\mathcal{Z}_r|$ and $m_s = |\mathcal{Z}_s|$. Thus the first ANN predicts secondary coordinates, not integration weights. Each additional ANN produces one scalar logit per element of its own fixed support. A prescribed output transformation converts these logits into positive element multipliers:

$$\begin{aligned} \mathbf{w}(\mathbf{q}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_r(\mathbf{q}; \boldsymbol{\theta}_r)), & \sum_{e \in \mathcal{Z}_r} w_e &= N_{\text{el}}, \\ \mathbf{v}(\mathbf{q}) &= N_{\text{el}} \text{softmax}(\mathcal{N}_s(\mathbf{q}; \boldsymbol{\theta}_s)), & \sum_{e \in \mathcal{Z}_s} v_e &= N_{\text{el}}. \end{aligned} \quad (39)$$

Here $\text{softmax}(\ell)_j = \exp(\ell_j) / \sum_k \exp(\ell_k)$ acts over the selected elements, not over training states. Input standardization is absorbed into each network map. As for the closure, the implementation evaluates these maps in the strain-informed coordinates $\boldsymbol{\xi} = \mathbf{A}_m^{-1} \mathbf{q}$. The softmax enforces positivity and the prescribed sum; integration accuracy must still be learned. The offline sequence is therefore closure training, support selection and pruning, then training of the two weight ANNs. Details of the pruning algorithm are given by Hernández et al. [32]; the following equations specify our neural weight-field fit.

Once a common support $\mathcal{Z} = \{e_1, \dots, e_m\}$ has been selected, weight-field training returns to the uncompressed contributions at each decoded training state. With the integrands defined above, the statewise system is

$$\mathbf{A}^s = \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix} \in \mathbb{R}^{(d+1) \times m}, \quad \mathbf{b}^s = \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix} \in \mathbb{R}^{d+1}. \quad (40)$$

Each vector $\mathbf{c}_{e_j}^s$ occupies d rows of one column. The first d rows reproduce the full-mesh integral; the last imposes the element-multiplier sum. In particular, $\mathbf{b}^s$ is computed from *all* elements, not by summing the selected columns of $\mathbf{A}^s$. Residual training uses $d = n$ and $\mathcal{Z} = \mathcal{Z}_r$; stress training uses $d = 4$ and its separate support $\mathcal{Z}_s$, replacing $\mathbf{w}$ by the stress multipliers $\mathbf{v}$.

For example, on a stress support of $m_s = |\mathcal{Z}_s|$ elements, the first four rows contain $p_{e,11}^s, p_{e,12}^s, p_{e,21}^s$, and $p_{e,22}^s$, respectively. Together with the row of ones, these form a $5 \times m_s$ matrix, denoted $\mathbf{A}_\sigma^s$ in Figure 2(c). Its target contains the four components of the full-mesh average $\bar{\mathbf{P}}^s = \sum_{e \in \mathcal{E}} \mathbf{p}_e^s$, followed by N_{el} . There are four stress rows rather than three because first Piola stress is not generally symmetric; conversion to the three-component second Piola output is performed afterwards, as described in Section 3.5. For fixed stress cubature, the same four-component contributions instead populate $\mathbf{M}_{\text{int}} \in \mathbb{R}^{N_{\text{el}} \times 4N_c}$ and follow the compression and selection in panel (a).

Unlike $\mathbf{G}_{\text{cub}}$, $\mathbf{A}^s$ contains physical component rows for a single state rather than compressed integration modes over the training set. Its column identities remain fixed as s changes, while both their contributions and the predicted weights vary; see Figure 2(b). For the residual-weight ANN, only $\boldsymbol{\theta}_r$ is optimized in the integration loss

$$\mathcal{L}_{\text{cub}}(\boldsymbol{\theta}_r) = \frac{1}{|\mathcal{I}_{\text{fit}}|} \sum_{s \in \mathcal{I}_{\text{fit}}} \frac{\|\mathbf{A}^s \mathbf{w}(\mathbf{q}^s) - \mathbf{b}^s\|_2^2}{\max(\|\mathbf{b}^s\|_2^2, \delta_{\text{floor}})}. \quad (41)$$

The dependence on $\boldsymbol{\theta}_r$ enters through $\mathbf{w}$ in Eq. (39); the sampled matrices and targets remain fixed. Stress-weight training optimizes $\boldsymbol{\theta}_s$ in the analogous loss using $\mathbf{A}_\sigma^s, \mathbf{b}_\sigma^s$, and $\mathbf{v}$. The small positive floor avoids division by zero. A preliminary fit to per-state constrained weights initializes each network; the final objective concerns integration error, not agreement with a unique target weight vector. Different weight vectors can reproduce the same integral. Their mean-square training loss is distinct from the median relative integration error used for checkpoint selection or reporting.

Online, a reduced equilibrium iterate supplies the same current state to the closure ANN and the residual-weight ANN: one reconstructs displacement, while the other supplies the multipliers used to assemble its projected residual. The stress-weight ANN supplies the separate output rule. Network parameters and support identities remain fixed; only their evaluated outputs change with the state. No pruning or network training is performed inside the equilibrium iterations. In particular, the iterative model evaluates weights at the current reduced state, not by imposing $\boldsymbol{\xi} = \mathbf{e}$; that prescription belongs to the direct mode introduced below.

Differentiation gives

$$\mathbf{r}_{\text{MAW}, \mathbf{q}} = \sum_{e \in \mathcal{Z}_r} \left[w_e \mathbf{B}_e^T \mathbf{K}_e \mathbf{B}_e + w_e \sum_j (f_{\text{int}, e})_j \partial_{\mathbf{q}\mathbf{q}}^2 (d_e)_j + \mathbf{h}_e (\nabla_{\mathbf{q}} w_e)^T \right]. \quad (42)$$

Here $\mathbf{K}_e = \partial_{\mathbf{d}_e} \mathbf{f}_{\text{int}, e}$. The three terms arise from element stiffness, decoder curvature, and weight variation, respectively. The microscopic iteration uses only the projected-stiffness term as a modified-Newton matrix, while refreshing decoder values, tangents, and weights at each iterate. This approximation concerns the iteration matrix, not the definition of equilibrium. The constitutive tangent must instead differentiate the complete residual and output, as described next.

For the affine HPROM with fixed weights, both additional terms vanish: the decoder has zero curvature and the weights have zero state derivative. Its projected stiffness is therefore the exact residual Jacobian. The iterative comparison consequently uses exact reduced Newton for the affine model and modified Newton for HPROM-ANN. Differences in iteration count and derivative cost are part of the measured implementation-level speedup, not an isolated measure of the advantage of nonlinear coordinates.

(a) Fixed ECM: all training states, one weight vector

$$\begin{array}{ccc}
 \begin{array}{c} \text{Truncated SVD} \\ \text{transpose} \\ \text{add constraint} \end{array} & & \begin{array}{c} \text{Greedy} \\ \text{selection} \end{array} \\
 \begin{array}{c} M_{\text{int}} = \begin{bmatrix} (\mathbf{c}_1^1)^T & \cdots & (\mathbf{c}_1^{N_c})^T \\ \vdots & & \vdots \\ (\mathbf{c}_{N_{\text{el}}}^1)^T & \cdots & (\mathbf{c}_{N_{\text{el}}}^{N_c})^T \end{bmatrix} \end{array} & \longrightarrow & \begin{array}{c} G_{\text{cub}} = \begin{bmatrix} U_{\text{cub}}^T \\ \mathbf{c}_{\perp}^T \end{bmatrix} \end{array} \\
 \begin{array}{c} \text{Rows: elements} \\ \text{Columns: state-component pairs} \\ N_{\text{el}} \times (dN_c) \end{array} & & \begin{array}{c} \text{Columns: candidate elements} \\ \text{Rows: modes} \\ \text{+ orthogonalized constant} \\ (r_c + 1) \times N_{\text{el}} \end{array} \\
 & & \begin{array}{c} G_{\text{cub}} \mathbf{w} \simeq \mathbf{b}_{\text{cub}}, \quad \mathbf{w} \geq \mathbf{0} \\ \mathbf{b}_{\text{cub}} = G_{\text{cub}} \mathbf{1} \end{array} \\
 & & \begin{array}{c} \text{One } \mathbf{w} \text{ for all retained modes} \\ w_e = 0 \text{ outside the support} \end{array}
 \end{array}$$

(b) MAW-ECM weight-field fit: one system per state, common support

$$\begin{array}{ccc}
 \begin{array}{c} \text{Columns: selected elements } e_1, \dots, e_m \\ \begin{bmatrix} \mathbf{c}_{e_1}^s & \cdots & \mathbf{c}_{e_m}^s \\ 1 & \cdots & 1 \end{bmatrix} \end{array} & \begin{array}{c} \begin{bmatrix} w_{e_1}(\mathbf{q}^s) \\ \vdots \\ w_{e_m}(\mathbf{q}^s) \end{bmatrix} \\ \underbrace{\hspace{1cm}}_{\mathbf{w}(\mathbf{q}^s)} \\ \text{State-dependent weights} \end{array} & \simeq & \begin{array}{c} \begin{bmatrix} \sum_{e \in \mathcal{E}} \mathbf{c}_e^s \\ N_{\text{el}} \end{bmatrix} \\ \mathbf{b}^s: \text{full-mesh sum} \\ (d+1) \times 1 \end{array} \\
 \begin{array}{c} \mathbf{A}^s: d \text{ component rows + normalization} \\ (d+1) \times m \end{array} & & & &
 \end{array}$$

$$A^1 \mathbf{w}(\mathbf{q}^1) \simeq \mathbf{b}^1$$

...

$$A^{N_c} \mathbf{w}(\mathbf{q}^{N_c}) \simeq \mathbf{b}^{N_c}$$

After pruning: one weight ANN shared across all training states

$$\mathbf{q} \xrightarrow{\mathcal{N}_r} \ell_r \xrightarrow{N_{\text{el}} \text{ softmax}} \mathbf{w}(\mathbf{q})$$

(c) Adaptive stress example of (b): four first Piola components + normalization

$$\begin{array}{ccc}
 \mathbf{p}_e^s = \frac{1}{|Y|} \sum_{g \in e} \omega_g \mathbf{P}_{\mu, g}^s, & \bar{\mathbf{P}}^s = \sum_{e \in \mathcal{E}} \mathbf{p}_e^s, & \mathcal{Z}_s = \{e_1, \dots, e_{m_s}\} \\
 \begin{bmatrix} p_{e_1, 11}^s & \cdots & p_{e_{m_s}, 11}^s \\ p_{e_1, 12}^s & \cdots & p_{e_{m_s}, 12}^s \\ p_{e_1, 21}^s & \cdots & p_{e_{m_s}, 21}^s \\ p_{e_1, 22}^s & \cdots & p_{e_{m_s}, 22}^s \\ 1 & \cdots & 1 \end{bmatrix} & \begin{array}{c} \mathbf{v}(\mathbf{q}^s) \\ N_{\text{el}} \text{ softmax}(\mathcal{N}_s(\mathbf{q}^s)) \end{array} & \simeq & \begin{bmatrix} \bar{P}_{11}^s \\ \bar{P}_{12}^s \\ \bar{P}_{21}^s \\ \bar{P}_{22}^s \\ N_{\text{el}} \end{bmatrix} \\
 \mathbf{A}_{\sigma}^s \in \mathbb{R}^{5 \times m_s}: \text{selected contributions} & & & \mathbf{b}_{\sigma}^s \in \mathbb{R}^5: \text{full-mesh target}
 \end{array}$$

Fixed stress ECM: stack these contributions in $M_{\text{int}} \in \mathbb{R}^{N_{\text{el}} \times 4N_c}$, then follow panel (a).

Figure 2: Construction of the cubature fitting systems. (a) Fixed ECM compresses the sampled integrands and fits one nonnegative sparse weight vector; its last row is an orthogonalized constant constraint. (b) Adaptive fitting uses uncompressed components, with one weight ANN and one support shared across states for each integration rule. (c) Stress instance of (b): four first Piola components and one normalization row, fitted with the stress multipliers $\mathbf{v}$. In (b)–(c), the scaled softmax enforces the last row by construction; the physical rows are fitted approximately. The prescribed sum N_{el} normalizes element multipliers, not physical volume. All targets use full-mesh sums. Residual and stress rules use separate supports and weight ANNs, distinct from the displacement-closure ANN.

3.5 Effective stress and consistent constitutive differentiation

Accurate integration of the projected residual does not automatically give accurate integration of the effective stress: these are different integrands. We therefore fit a separate output support $\mathcal{Z}_s$ and multipliers v_e , using fixed ECM for HPROM or the adaptive procedure of Section 3.4.2 for HPROM-ANN. In the following common expression, v_e is constant in the former case and depends on the reduced state in the latter. With microscopic first Piola stress $\mathbf{P}_{\mu,g}$ and reference quadrature measure ω_g , the stress reconstruction is

$$\bar{\mathbf{P}}_h(\mathbf{q}, \mathbf{e}) = \frac{1}{|Y|} \sum_{e \in \mathcal{Z}_s} v_e(\mathbf{q}) \sum_{g \in e} \omega_g \mathbf{P}_{\mu,g}(\mathbf{d}_e), \quad \bar{\mathbf{S}}_h = \bar{\mathbf{F}}^{-1} \bar{\mathbf{P}}_h. \quad (43)$$

Equation (43) provides the output counterpart of the residual rules in Eqs. (31) and (37). The residual and output supports may overlap. Their union specifies the element data required by an iterative constitutive evaluation, but only the residual support is used in equilibrium iterations.

The implemented three-component output takes the symmetric part of the converted tensor:

$$\mathbf{s}_h(\mathbf{q}, \mathbf{e}) = ((\bar{\mathbf{S}}_h)_{11}, (\bar{\mathbf{S}}_h)_{22}, \tfrac{1}{2}[(\bar{\mathbf{S}}_h)_{12} + (\bar{\mathbf{S}}_h)_{21}])^T. \quad (44)$$

This states the actual output convention without assuming that an approximate displacement and independently fitted stress cubature give an exactly symmetric raw tensor. The constitutive derivative refers to this output, including the symmetrization.

Once reduced equilibrium determines $\mathbf{q}^*(\mathbf{e})$, the effective map is $\mathbf{s}_h(\mathbf{q}^*(\mathbf{e}), \mathbf{e})$. If its residual Jacobian is nonsingular, the same implicit differentiation as in Section 2 gives

$$\mathbf{D}_h = \mathbf{s}_{h,e} - \mathbf{s}_{h,q} \mathbf{r}_{h,q}^{-1} \mathbf{r}_{h,e}. \quad (45)$$

Partial derivatives hold the other argument fixed and are evaluated at the converged state. In particular, $\mathbf{s}_{h,e}$ includes the periodic lifting and the first-to-second-Piola conversion. The residual derivative must be the derivative of the full residual, not merely the modified-Newton matrix used to reach equilibrium.

The implementations use the same implicit-function formula but divide its evaluation differently between analytic and numerical derivatives. For the affine HPROM, the assembled reduced stiffness and analytic lifting derivative give $d\mathbf{q}_{\text{tra}}^*/d\mathbf{e}$ through three linear right-hand sides. Central differences are then applied only to the stress, along these linearized equilibrium directions. For HPROM-ANN, the implementation instead uses central differences at fixed $\mathbf{q}$ or fixed $\mathbf{e}$ for all four partial-derivative matrices in Eq. (45), including the effects of decoder curvature and weight variation; a small linear solve combines them explicitly. These derivatives could also be obtained analytically or by automatic differentiation: numerical differentiation is an implementation choice, not a limitation of the formulation. Neither procedure re-solves nonlinear microscopic equilibrium at every perturbed strain, and both remain subject to differentiation and equilibrium-solver tolerances.

3.6 Direct decoder evaluation: D-HPROM-ANN

When input-informed coordinates closely approximate the loading parameters, the decoder can also be evaluated without solving reduced equilibrium, as discussed in the hyperelastic example of Ref. [32]. Here this defines a separate operating mode, D-HPROM-ANN, obtained by prescribing $\boldsymbol{\xi} = \mathbf{e}$ in Eq. (27):

$$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e}). \quad (46)$$

The decoded state is inserted directly into the adaptive stress rule, using the same closure and stress-weight field as the iterative model. Only the stress support is needed; no residual cubature or reduced equilibrium Jacobian is evaluated. The tangent is obtained by central differences of the complete direct stress map, updating the decoder, weights, lifting, and stress conversion at each perturbed input.

This route still evaluates the microscopic material law on selected elements; it is not a neural stress regressor or a learned macroscopic potential. The prefix D is our label for the direct operating mode, not a new projection method. Its accuracy depends on both the closure and the approximation $\boldsymbol{\xi} \approx \mathbf{e}$. Conversely, solving projected equilibrium corrects the primary state only within the chosen manifold and cubature approximation, so it does not guarantee a smaller stress error than direct evaluation at every input.

3.7 Retained mechanical structure and its limits

All of these reduced models evaluate the original microscopic hyperelastic law, and the iterative variants additionally impose projected equilibrium. These retained ingredients distinguish them from direct constitutive regression, but do not by themselves certify an effective energy potential.

With fixed weights and consistent kinematics, Galerkin forces derived from element energies are the gradient of a weighted reduced energy. This is the variational basis for structure-preservation results for ECSW under their stated hypotheses [28]. Transferring such a conclusion to the deployed effective stress map requires two additional distinctions.

First, the independently fitted equilibrium and stress cubatures need not define derivatives of one energy. Second, for element energies Π_e , weighting their derivatives differs from differentiating an energy with state-dependent weights:

$$\nabla_{\mathbf{q}} \sum_e w_e(\mathbf{q}) \Pi_e(\mathbf{q}, \mathbf{e}) = \sum_e w_e(\mathbf{q}) \nabla_{\mathbf{q}} \Pi_e + \sum_e \Pi_e \nabla_{\mathbf{q}} w_e. \quad (47)$$

The last term is absent from Eq. (37), and positivity of the weights does not remove it. Thus the weighted sum of element energies is not, in general, the potential of that residual. These observations do not assert failure at every state; they delimit what the construction guarantees. We assign neither an exact effective potential nor a polyconvexity certificate to the deployed HPROM variants without further analysis. A consistent tangent differentiates the implemented stress map; it does not make that map conservative or guarantee tangent symmetry.

3.8 Offline–online separation and reduced meshes

The offline stage collects HDM snapshots, constructs the ROBs and latent coordinates, fits the closure, selects cubature supports, trains adaptive weights where applicable, and builds reduced meshes. The online stage evaluates the prescribed lifting, reconstructs only the displacement entries required by the active supports, solves reduced equilibrium when needed, and returns effective stress and tangent.

An optimized reduced mesh is constructed from the selected elements and their required nodes, preserving the original geometry, material law, and quadrature. The basis operators and periodic lifting, including its derivatives, are restricted to this support before online evaluation; the optimized implementations do not first construct a full-dimensional lifting vector. Cubature multipliers account approximately for omitted contributions. Element forces and stiffnesses are projected locally, and derivative evaluations are batched where possible. The direct mode also omits decoder derivatives unnecessary for stress evaluation.

Consequently, online cost depends on support size, retained secondary dimension, equilibrium iterations, and derivative evaluation, not only on the number of primary coordinates. Reducing the equilibrium dimension alone therefore does not determine the attainable speedup. Table 1 summarizes the deployed reduced models and their HDM reference; the full-mesh PROM constructions serve as intermediate formulations.

Table 1: Deployed reduced evaluators and their HDM reference. The independent displacement representations below are lifted to physical nodal fields through Eq. (13). Stress is reconstructed from microscopic constitutive evaluations.

Model	State representation	Equilibrium unknowns	Integration	Neural prediction
HDM / FOM	$\mathbf{u} \in \mathbb{R}^N$	N	Full mesh	None
HPROM	$\tilde{\mathbf{u}} = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tra}} \mathbf{q}_{\text{tra}}$	n_{tra}	Fixed ECM	None
HPROM-ANN	$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}(\mathbf{q})$	n	Adaptive ECM	Secondary coordinates and weights
D-HPROM-ANN	$\tilde{\mathbf{u}}_D(\mathbf{e}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{A}_m \mathbf{e} + \bar{\mathbf{V}} \hat{\mathcal{N}}(\mathbf{e})$	None	Adaptive stress rule	Same closure and stress weights

4 Learned constitutive laws

The models of Section 3 reconstruct microscopic displacements and evaluate the constituent law. The non-intrusive route instead learns the effective constitutive map itself, without reconstructing the RVE state online. Given pairs of macroscopic strains and effective stresses, the most direct approach is to train a neural network to predict the three stress components from $\mathbf{e}$. We call this model *Regression*. Choosing objective strain inputs and a symmetric material-stress output preserves the corresponding transformation properties. It does not, however, ensure that the predicted stresses derive from a stored energy: fitting their values at sampled states does not enforce compatibility between their cross derivatives. The fitted law can therefore

perform nonzero net work around a closed strain cycle, even when its training data come from a hyperelastic material [7].

A natural way to remove this inconsistency is to learn one scalar energy and obtain every stress component by differentiation, $\mathbf{s} = \nabla_{\mathbf{e}} W$. The stress predictions are then linked by a common potential rather than fitted as independent outputs. As’ad et al. [7] use this distinction to introduce a hyperelastic ANN alongside direct stress regression. We adopt the same potential-based principle for an energy baseline termed *Free*: the name indicates that no convexity constraint is imposed on its energy network, not that the model is devoid of mechanical structure. Its objective inputs and reference normalization retain a stress-free, zero-energy reference state, while energy differentiation enforces a conservative constitutive response.

Learning a potential leaves a further question unanswered: what restrictions should be placed on its shape? A smooth energy alone does not ensure rank-one convexity or prevent finite-energy volume collapse. As’ad et al. [7] also consider an energy convex in the Green–Lagrange strain. Here we pursue a different condition, polyconvexity, by constructing an energy convex in independent deformation minors. The distinction matters at finite strain and is made precise below. We realize this construction with two alternative cores, an input-convex neural network (ICNN) and an input-convex Kolmogorov–Arnold network (ICKAN), coupled to the same directional-feature family and analytic normalization and growth terms.

The two baselines isolate different constitutive choices. We use the following short names consistently in the text, figures, and tables:

- **Regression: direct stress regression.** The network predicts three stress components from $\mathbf{e}$, without imposing a common energy potential. Objective inputs and a symmetric stress output are retained.
- **Free: hyperelastic energy without convexity constraints.** The network represents one scalar potential, and stresses follow by differentiation. The energy is normalized to zero energy and stress at the reference state; “Free” refers specifically to the absence of convexity constraints.

ICNN and ICKAN complete the comparison as two realizations of the proposed polyconvex energy. To specify what each choice guarantees, we first state the mechanical criteria and distinguish potential structure from finite-strain stability. We then detail the baselines, construct the features and cores, and establish the properties of the complete energy. Throughout this section, $\mathbf{F} \in \text{GL}^+(2)$ and $J = \det \mathbf{F} > 0$. Derivatives with respect to deformation are taken with the learned parameters fixed.

4.1 Admissibility criteria and scope of the guarantees

A small prediction error is not sufficient to define an admissible hyperelastic surrogate. We organize the construction around six criteria, following the distinctions made in constitutive PANN formulations [8, 9]. The enumeration is an organizing device for this paper, not a universal checklist or a claim that polyconvexity is necessary for every physical material:

- (C1) *Potential structure*: stress is the derivative of a single-valued stored energy in work-conjugate variables.
- (C2) *Objectivity*: a superposed rigid rotation does not change that energy.
- (C3) *Stress symmetry and anisotropy*: the material stress is symmetric, while directional dependence is allowed without imposing an unverified material symmetry group.
- (C4) *Reference normalization*: the undeformed state has zero energy and zero stress.
- (C5) *Polyconvexity*: the energy admits a convex representation in deformation minors, providing a sufficient rank-one convexity condition.
- (C6) *Growth*: energy diverges under volume collapse, unbounded volume expansion, and unbounded distortion at bounded positive volume.

The ICNN and ICKAN constructions below satisfy these criteria for every finite parameter set obeying their architectural constraints. Global nonnegative energy is an additional condition checked separately for the selected parameters. Constitutive identities, sufficient bounds, and sampled numerical checks consequently play different roles.

Potential structure (C1). For a differentiable hyperelastic energy, stress is its derivative in work-conjugate variables. In the convention of Eq. (2),

$$\mathbf{s}(\mathbf{e}) = \nabla_{\mathbf{e}} W(\mathbf{e}), \quad \oint_{\Gamma} \mathbf{s} \cdot d\mathbf{e} = 0. \quad (48)$$

For the isothermal purely elastic setting considered here, (C1) also gives $\mathbf{s} \cdot \dot{\mathbf{e}} - \dot{W} = 0$: the mechanical dissipation vanishes. The closed-path identity is exact for a single-valued differentiable potential. On a simply connected strain domain, a continuously differentiable stress map has such a potential if its Jacobian is symmetric. Direct stress regression need not impose the cross-derivative identities $\partial s_i / \partial e_j = \partial s_j / \partial e_i$ [7]. A small pointwise stress error alone does not establish them.

Objectivity and the three meanings of symmetry (C2)–(C3). For a superposed proper rotation $\mathbf{Q}$, objectivity requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}). \quad (49)$$

Using $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as an input to an energy enforces the first identity; differentiating the energy then gives the stress transformation. A direct map from $\mathbf{E}$ to the material stress $\mathbf{S}$ can also be objective without possessing a potential. Objectivity therefore does not distinguish Regression from Free in this study. The symmetry $\mathbf{S} = \mathbf{S}^T$ follows from differentiation with respect to symmetric strain for the energy models and is built into the three-component output of Regression. It implies a symmetric Cauchy stress $\boldsymbol{\sigma} = J^{-1} \mathbf{F} \mathbf{S} \mathbf{F}^T$, as required by angular-momentum balance in the present continuum.

Neither stress symmetry nor objectivity imposes a material symmetry group. A right action $\mathbf{F} \mapsto \mathbf{F} \mathbf{Q}_m$ leaves the energy unchanged only when $\mathbf{Q}_m$ belongs to that group. Directional features are fixed in the reference material and are unchanged by a superposed spatial rotation. Finally, the constitutive tangent $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e} \in \mathbb{R}^{3 \times 3}$ measures how the three stress components change with engineering strain, with entries $D_{ij} = \partial s_i / \partial e_j$. For an energy-based law it is the Hessian $\partial_{\mathbf{e}\mathbf{e}}^2 W$. Its *major symmetry*, $\mathbf{D} = \mathbf{D}^T$, is a third property: it follows from a twice differentiable scalar energy, not merely from a symmetric stress tensor.

Reference normalization (C4). Reference normalization imposes

$$W(\mathbf{I}) = 0, \quad \mathbf{P}(\mathbf{I}) = \mathbf{0}. \quad (50)$$

The first condition fixes an arbitrary additive energy constant. The second is a substantive stress condition and depends on how the energy is corrected. Subtracting a linear function of $\mathbf{E}$ preserves a potential but need not preserve polyconvexity in $\mathbf{F}$. An affine correction in independent deformation minors, in contrast, preserves convexity in those variables. This is the representation introduced next and used by our normalization.

Polyconvexity and finite-strain stability (C5). Potential structure does not impose a stability condition. The Legendre–Hadamard condition for a twice differentiable energy is

$$\partial_{\mathbf{F}\mathbf{F}}^2 W(\mathbf{F})[\mathbf{a} \otimes \mathbf{b}, \mathbf{a} \otimes \mathbf{b}] \geq 0 \quad \text{for all } \mathbf{a}, \mathbf{b}. \quad (51)$$

Here $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ are arbitrary vectors, $(\mathbf{a} \otimes \mathbf{b})_{ij} = a_i b_j$ is their outer product, and $t \in \mathbb{R}$ parametrizes the perturbation. The condition tests the curvature of the scalar function $t \mapsto W(\mathbf{F} + t \mathbf{a} \otimes \mathbf{b})$ at $t = 0$: the energy must not curve downward along a rank-one perturbation. Strict positivity for nonzero directions is a stronger statement than this nonnegative condition. In two dimensions, a sufficient condition is a representation

$$W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F}), \quad \mathcal{P} \text{ convex in its independent arguments.} \quad (52)$$

Here $\mathcal{P}$ is defined on the convex set $\mathbb{R}^{2 \times 2} \times (0, \infty)$, treating its matrix and scalar arguments as independent; only its evaluation in W sets the second argument equal to $\det \mathbf{F}$. This distinction is essential: we do not require W to be convex in $\mathbf{F}$ itself. The cofactor is linear in the entries of a 2×2 matrix, so it need not be introduced as an additional independent argument. Along a rank-one perturbation, both $\mathbf{F}$ and $\det \mathbf{F}$ vary affinely with the perturbation parameter. The resulting path is therefore a straight line in the arguments of $\mathcal{P}$, whose convexity gives Eq. (51). Appendix A supplies the derivation. Testing finitely many rank-one directions cannot establish polyconvexity.

The standard three-dimensional definition uses $(\mathbf{F}, \text{cof } \mathbf{F}, \det \mathbf{F})$ [12, 13]. The guarantee established here is two-dimensional; it does not establish an admissible three-dimensional extension with arbitrary out-of-plane deformation.

Convexity in $\mathbf{E}$ is a different property because $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I})$ depends quadratically on $\mathbf{F}$. Consequently, the energy curvature along a linear perturbation of $\mathbf{F}$ includes both a strain-tangent contribution and a

contribution from the curvature of the strain path, weighted by the current stress. The latter can be negative in compression, so a positive-semidefinite strain Hessian $\mathbf{D}$ alone does not guarantee Eq. (51) [7, 10]. Convexity in $\mathbf{C}$ and in $\mathbf{E}$ are equivalent up to their affine change of variables, but neither implies polyconvexity. Appendix A derives this stress-dependent curvature and gives an analytical counterexample.

Growth and the limits of the guarantees (C6). Finally, the inequalities $W(\mathbf{F}) \geq 0$, the barrier $W \rightarrow \infty$ as $J \rightarrow 0^+$, and the existence of minimizers are not synonyms. Ciarlet’s presentation of Ball’s existence argument [13], especially Sections 4.9 and 7.7, includes coercivity, admissible boundary data, and finite-energy competitors in addition to polyconvexity. Neither the existence theory nor a constitutive barrier guarantees convergence of an unconstrained Newton step. In this work, architectural statements, saved-parameter bounds, and numerical observations are therefore identified separately.

4.2 Direct stress and unconstrained-energy baselines

Regression directly approximates $\mathbf{e} \mapsto \mathbf{s}$ with three stress outputs. The Free model instead uses the objective input vector $(C_{11} - 1, C_{22} - 1, C_{12}, J - 1)^T$ and returns one scalar energy. Since $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $J = \sqrt{\det \mathbf{C}}$, these four inputs are functions of the same three strain components. Writing the resulting composite map as $\widetilde{W}(\mathbf{e})$, we impose the reference conditions through

$$W_{\text{free}}(\mathbf{e}) = \widetilde{W}(\mathbf{e}) - \widetilde{W}(\mathbf{0}) - \partial_{\mathbf{e}} \widetilde{W}(\mathbf{0}) \cdot \mathbf{e}. \quad (53)$$

This enforces zero reference energy and stress without enforcing polyconvexity. Using material-coordinate components of $\mathbf{C}$ does not impose isotropy. Reference-axis changes must transform the material directions together with the tensor components.

4.3 Feature design: expressiveness and admissibility

The directional construction can be understood as a continuation of invariant-based constitutive modeling. For a three-dimensional right Cauchy–Green tensor $\mathbf{C}_3$, the familiar first, second, and third principal invariants are

$$I_1^{(3)} = \text{tr } \mathbf{C}_3, \quad I_2^{(3)} = \frac{1}{2} [(\text{tr } \mathbf{C}_3)^2 - \text{tr}(\mathbf{C}_3^2)], \quad I_3^{(3)} = \det \mathbf{C}_3. \quad (54)$$

They describe principal stretches without recording their orientation relative to the material. An energy depending only on these scalars is therefore isotropic. In the present two-dimensional setting, the corresponding two independent principal invariants are $I_1 = \text{tr } \mathbf{C}$ and $\det \mathbf{C} = J^2$; a third independent principal invariant is not available.

To retain directional information, one can supplement the isotropic measures with a structural direction $\mathbf{d}$. The familiar anisotropic stretch invariant is

$$I_4(\mathbf{d}) = \mathbf{d}^T \mathbf{C} \mathbf{d} = \mathbf{C} : (\mathbf{d} \otimes \mathbf{d}) = \|\mathbf{F} \mathbf{d}\|^2, \quad \|\mathbf{d}\| = 1. \quad (55)$$

It measures squared stretch along a material direction and remains unchanged by a superposed spatial rotation. Such directional invariants are standard ingredients of anisotropic hyperelastic energies [33, 34]. Here $\mathbf{d}$ need not represent a physical fiber: learned directions provide objective scalar descriptors of the RVE’s effective anisotropy. For an orthonormal pair $(\mathbf{d}, \mathbf{d}^\perp)$ in two dimensions,

$$I_4(\mathbf{d}) + I_4(\mathbf{d}^\perp) = I_1, \quad \|\text{cof } \mathbf{F} \mathbf{d}\|^2 = I_4(\mathbf{d}^\perp). \quad (56)$$

The second identity also connects cofactor-based directional measures to this same in-plane stretch description.

Consider a feature vector $\mathbf{z}(\mathbf{F})$ and a scalar core H nondecreasing in each input. For any two deformation states,

$$\mathbf{z}(\mathbf{F}_B) \geq \mathbf{z}(\mathbf{F}_A) \text{ componentwise} \implies H(\mathbf{z}(\mathbf{F}_B)) \geq H(\mathbf{z}(\mathbf{F}_A)). \quad (57)$$

This implication holds regardless of the network’s width. If the complete constitutive energy is $W = H(\mathbf{z}) + W_{\text{corr}}$, where W_{corr} collects the analytic normalization and growth terms, the ordered quantity is $W - W_{\text{corr}}$, not W alone. Thus a decreasing target energy along such a path indicates an obstruction only after the complete correction is accounted for. The diagnosis must examine the feature map and analytic terms together.

This observation explains why the choice of an admissible feature construction can matter more than the capacity of its convex core. Changing the number of features does not change the number of independent strains: the features remain functions of the same three engineering-strain components. They embed that

state into variables on which the network is constrained to be monotone and convex. Additional features can enlarge the admissible response family, but their number alone does not prove completeness.

The paired construction below couples directional stretch to volume through admissible determinant exponents. A feature can therefore decrease along a path when its denominator grows sufficiently, even if a numerator increases. Moreover, its reference derivative is controlled analytically, allowing normalization and volumetric stiffness to be set independently.

4.3.1 Paired directional features

Let m denote the number of scalar features. For each $i = 1, \dots, m$, let $\mathbf{d}_i = (\cos \theta_i, \sin \theta_i)^T$ and $\mathbf{d}_i^\perp = (-\sin \theta_i, \cos \theta_i)^T$. Define

$$\begin{aligned} z_i(\mathbf{F}, J) &= \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} + \frac{I_4(\mathbf{d}_i^\perp)^{q_i}}{q_i J^{c_i}} \\ &= \frac{\|\mathbf{F} \mathbf{d}_i\|^{2p_i}}{p_i J^{b_i}} + \frac{\|\mathbf{F} \mathbf{d}_i^\perp\|^{2q_i}}{q_i J^{c_i}}, \end{aligned} \quad (58)$$

where θ_i sets the orientation of the reference directions, p_i, q_i are the powers of the squared-stretch invariants, and b_i, c_i control each contribution's coupling to the area ratio J . The exponents satisfy

$$p_i, q_i \geq \frac{1}{2}, \quad 0 \leq b_i \leq 2p_i - 1, \quad 0 \leq c_i \leq 2q_i - 1. \quad (59)$$

The parameters $\theta_i, p_i, q_i, b_i, c_i$ are learned subject to Eq. (59). We use $\text{softplus}(x) = \log(1 + e^x)$ and $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$, whose ranges for $x \in \mathbb{R}$ are $(0, \infty)$ and $(0, 1)$, respectively. With unconstrained parameters $\hat{p}_i, \hat{q}_i, \hat{b}_i, \hat{c}_i \in \mathbb{R}$, set

$$\begin{aligned} p_i &= \frac{1}{2} + \text{softplus}(\hat{p}_i), & b_i &= (2p_i - 1) \text{sigmoid}(\hat{b}_i), \\ q_i &= \frac{1}{2} + \text{softplus}(\hat{q}_i), & c_i &= (2q_i - 1) \text{sigmoid}(\hat{c}_i). \end{aligned} \quad (60)$$

The angles are unconstrained. These transformations preserve the inequalities during training; a penalty loss is not needed to enforce them. For finite real values of the unconstrained parameters, the bounds in Eq. (59) hold strictly; boundary values are approached as limits. Each z_i is one scalar feature containing two perpendicular contributions. The two contributions need not have equal exponents. Consequently, pairing perpendicular directions does not impose invariance under their interchange. In the numerical example of Section 5, both cores use $m = 32$ paired features.

Besides ensuring convexity, the paired construction makes the reference strain gradient of each feature isotropic. Consequently, the neural contribution has an isotropic reference stress, which a single scalar determinant correction can cancel in Section 4.5. The following result records the properties needed for that construction.

Proposition 1. *Under the constraints in Eq. (59), each feature in Eq. (58) is objective and convex in the independent variables $(\mathbf{F}, J)$ on $J > 0$. Its value and engineering-strain gradient at the identity, evaluated with $J = \det \mathbf{F}$, are*

$$z_i^0 = \frac{1}{p_i} + \frac{1}{q_i}, \quad \partial_{\mathbf{e}} z_i|_{\mathbf{0}} = \rho_i(1, 1, 0)^T, \quad \rho_i = 2 - \frac{b_i}{p_i} - \frac{c_i}{q_i}. \quad (61)$$

The proof is given in Appendix B. The mechanism behind the reference gradient is that $1/p_i$ and $1/q_i$ cancel the stretch-power multipliers upon differentiation. The two perpendicular directional contributions then sum to $2\mathbf{I}$, while the determinant terms contribute $-(b_i/p_i + c_i/q_i)\mathbf{I}$. Thus ρ_i is the scalar amplitude of an isotropic reference gradient: its two normal components are equal and its shear component is zero. This reference property does not imply isotropy of the response away from the reference state.

We condition the core inputs using $y_i = (z_i - z_i^{\text{cen}})/s_i$, where z_i^{cen} is a centering value and $s_i > 0$ is a scaling factor. Both are independent of deformation for any fixed parameter set. This positive affine transformation preserves convexity in independent $(\mathbf{F}, J)$. The choices used in the numerical example are specified in Section 5.2.

Learning the directions does not amount to identifying or enforcing a complete material symmetry group. No completeness or universal-approximation theorem is asserted for a finite collection of these features followed by a monotone convex core.

4.4 ICNN and integrated-hat spline cores

The feature proposition is only the first part of the construction. Its composition with a neural core is convex in the independent minors if that core is both convex and componentwise nondecreasing [8, 9]. An input-convex network without this additional monotonicity need not preserve the convexity of nonlinear features.

Figure 3 contrasts where the two cores place their learned transformations.

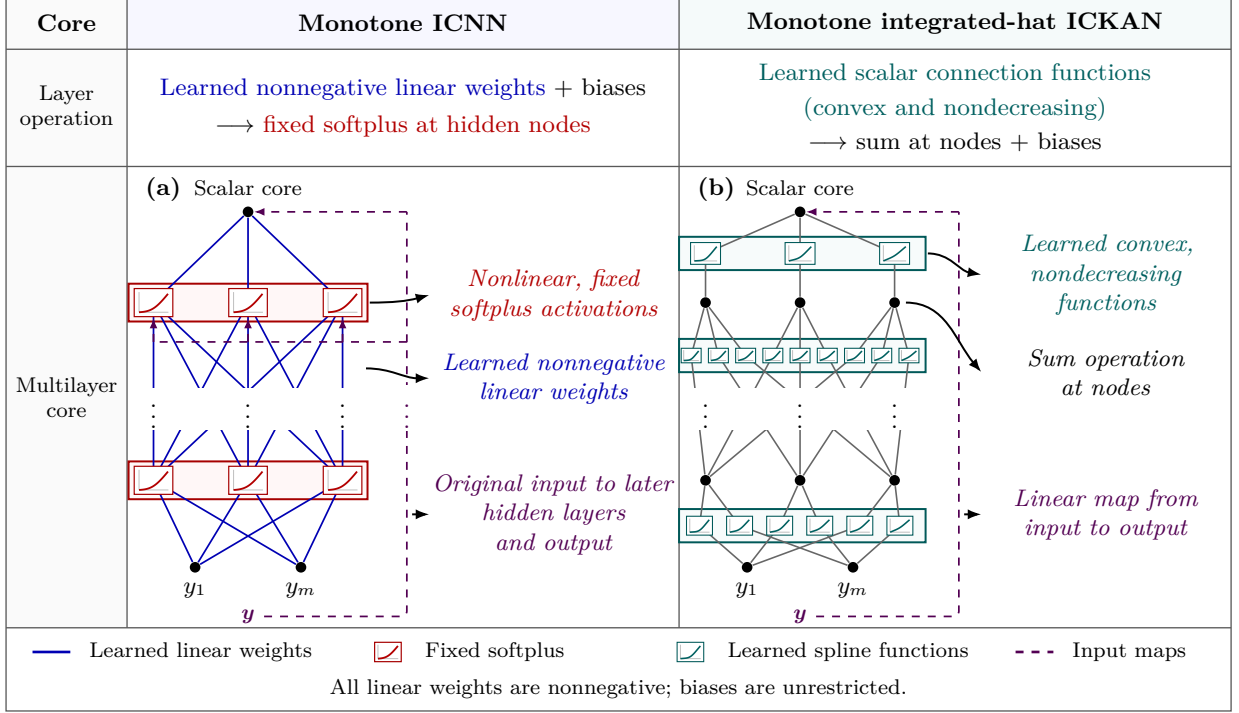


Figure 3: General multilayer structure of the monotone ICNN (a) and integrated-hat ICKAN (b). Blue connections carry nonnegative learned linear weights; red boxes apply fixed softplus; green boxes contain independently learned convex, nondecreasing spline functions. ICNN hidden nodes apply softplus after an affine sum, while the output is linear. ICKAN nodes sum transformed contributions. Purple dashed paths bundle linear maps of the entire conditioned input $\mathbf{y}$, with nonnegative weights: the ICNN supplies original-input contributions to every hidden layer and the output; the ICKAN’s additional input map goes only to the output. First-layer ICNN input weights are shown in blue. Ellipses denote intervening hidden layers; widths and curves are illustrative. Unrestricted biases are omitted. The diagram depicts the neural cores, not the complete energy.

4.4.1 Monotone ICNN

Let L denote the number of hidden layers and $\mathbf{a}^{(\ell)}$ the vector of activations in layer ℓ , for $\ell = 1, \dots, L$. We write $\hat{H}(\mathbf{y})$ for the scalar neural core acting on the conditioned features $\mathbf{y}$; the complete energy additionally includes the reference-normalization and growth terms of Section 4.5. The deployed ICNN has the form

$$\begin{aligned}
 \mathbf{a}^{(1)} &= \phi(\mathbf{U}_1 \mathbf{y} + \mathbf{b}_1), \\
 \mathbf{a}^{(\ell)} &= \phi(\mathbf{W}_\ell \mathbf{a}^{(\ell-1)} + \mathbf{U}_\ell \mathbf{y} + \mathbf{b}_\ell), \quad \ell = 2, \dots, L, \\
 \hat{H}(\mathbf{y}) &= \mathbf{v}^T \mathbf{a}^{(L)} + \mathbf{w}^T \mathbf{y} + b_{\text{out}},
 \end{aligned} \tag{62}$$

where $\phi = \text{softplus}$ is a smooth, convex, and increasing activation applied componentwise. The matrices $\mathbf{U}_\ell$ and $\mathbf{W}_\ell$ contain the weights from $\mathbf{y}$ and the preceding hidden layer, respectively, and $\mathbf{b}_\ell$ is the layer bias vector. The output weight vectors $\mathbf{v}$ and $\mathbf{w}$ act on $\mathbf{a}^{(L)}$ and $\mathbf{y}$, respectively; b_{out} is an additive scalar output bias.

Every entry of $\mathbf{W}_\ell, \mathbf{U}_\ell, \mathbf{v}, \mathbf{w}$ is nonnegative. Softplus-based transformations of unconstrained weight parameters impose these signs. Biases are unrestricted because adding constants to the layer preactivations preserves convexity and monotonicity. Hidden biases can change the core’s derivatives with respect to $\mathbf{y}$, whereas b_{out} only shifts $\hat{H}$ by a constant and leaves these derivatives unchanged.

The first hidden layer is convex and nondecreasing in $\mathbf{y}$. A nonnegative weighted sum, addition of an affine input map with nonnegative linear coefficients, and composition with convex nondecreasing ϕ preserve both properties at every subsequent layer and at the output. This is the ICNN composition argument of Amos et al. [35], with the additional nonnegativity constraints on $\mathbf{U}_\ell$ and $\mathbf{w}$ imposed here to guarantee monotonicity. Convexity in $\mathbf{y}$ alone does not require the latter restriction.

4.4.2 Monotone integrated-hat ICKAN

Unlike the ICNN, which applies a fixed activation after an affine weighted sum at each hidden node, the ICKAN assigns a learned scalar function to each connection (edge) between layers. Each node sums the transformed contributions from its incoming connections. For an individual edge, let x denote its scalar input: a component of $\mathbf{y}$ in the first layer, or an output of the preceding layer thereafter. The edge function has the form

$$f(x) = ax + \sum_k c_k B_k(x) + d, \quad a, c_k \geq 0, \quad (63)$$

where a is the learned linear slope, c_k are learned weights of the spline bases B_k , k indexes these bases, and d is an unrestricted additive constant. Softplus transformations of unconstrained parameters impose the signs of a and c_k . In the implementation, additive constants are collected into one trainable bias per output node.

Each B_k'' is a nonnegative triangular hat centered at a knot κ_k . The knots form a fixed uniform grid with spacing $h > 0$; neither their locations nor the spacing are optimized. The grid used in the numerical example is specified in Section 5.2. Writing $s = (x - \kappa_k)/h$, the curvature is

$$B_k''(x) = \max(1 - |s|, 0). \quad (64)$$

Integrating this hat twice, with $B_k(x) = B_k'(x) = 0$ for $x \leq \kappa_k - h$, gives the integrated-hat basis

$$B_k(x) = h^2 \begin{cases} 0, & s \leq -1, \\ (s+1)^3/6, & -1 < s \leq 0, \\ (-s^3 + 3s^2 + 3s + 1)/6, & 0 < s < 1, \\ s, & s \geq 1. \end{cases} \quad (65)$$

The curvature is localized to $[\kappa_k - h, \kappa_k + h]$, not the entire basis: B_k has a zero left tail and a linear right tail.

Here $B_k'(-\infty) = 0$ and $B_k'' \geq 0$, so $B_k' \geq 0$. It follows that $f' \geq 0$ and $f'' \geq 0$, with continuous second derivative. Each layer sums scalar edges applied to its input coordinates; composing such layers preserves convexity and componentwise monotonicity. The deployed core also includes a linear input-to-output skip from $\mathbf{y}$ with nonnegative weights, which preserves the same properties.

In Eq. (65), values and the first two derivatives match analytically at every join, and the curvature vanishes at the outer joins. Each scalar edge is therefore globally C^2 with linear tails, while its curvature can decrease to zero without becoming negative. These tails are linear in the scalar input x ; they do not imply a linear energy in $\mathbf{F}$, because the complete construction also includes nonlinear features and analytic growth terms. This core is inspired by the monotonic input-convex KAN framework [41], using the integrated-hat bases and analytic tail construction specified here.

4.5 The complete energy and its admissibility guarantees

We construct the effective stored energy by combining the neural contribution with reference-normalization and analytic growth terms:

$$\begin{aligned} W(\mathbf{F}) = & \underbrace{H(\mathbf{z}) - H(\mathbf{z}^0)}_{(i)} + \underbrace{[-r(J-1)]}_{(ii)} \\ & + \underbrace{\alpha(J-1 - \log J) + \frac{\beta}{2}(J-1)^2}_{(iii)} \\ & + \underbrace{\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right)}_{(iv)}, \quad \alpha, \beta, \varepsilon > 0. \end{aligned} \quad (66)$$

Here $J = \det \mathbf{F}$, $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $\mathbf{z} = \mathbf{z}(\mathbf{F}, J)$ is the physical feature vector. The positive energy-density coefficients α , β , and ε weight the analytic growth terms. We write $H(\mathbf{z}) = \hat{H}(\mathbf{y}(\mathbf{z}))$ for the neural

contribution evaluated through the prescribed centering and positive scaling of the features. For either core, this composition is twice continuously differentiable, convex, and componentwise nondecreasing in $\mathbf{z}$.

Let $\mathbf{z}^0 = \mathbf{z}(\mathbf{I}, 1)$ denote the reference feature vector defined in Eq. (61). The reference sensitivities of the neural contribution and the resulting scalar stress-correction coefficient are

$$h_i = \partial_{z_i} H(\mathbf{z}^0) \geq 0, \quad r = \sum_i h_i \rho_i. \quad (67)$$

where ρ_i are the reference feature-gradient factors in Eq. (61). Both $H(\mathbf{z}^0)$ and r are computed from the current model parameters. They can change during training, but remain constant when differentiating with respect to deformation at fixed parameters.

The groups marked (i)–(iv) in Eq. (66) have the following roles:

- (i) *Setting the energy reference.* At $\mathbf{F} = \mathbf{I}$, the features equal $\mathbf{z}^0$, $J = 1$, and $\text{tr } \mathbf{C} = 2$. Every term added to the neural contribution then vanishes except the constant subtraction, giving

$$W(\mathbf{I}) = H(\mathbf{z}^0) - H(\mathbf{z}^0) = 0. \quad (68)$$

Since $H(\mathbf{z}^0)$ is constant with respect to deformation, subtracting it leaves the stress and tangent unchanged.

- (ii) *Removing the reference stress.* Stress is the derivative of energy with respect to the work-conjugate strain $\mathbf{e}$. By the chain rule and Eq. (61), the neural contribution has reference stress

$$\nabla_{\mathbf{e}} H(\mathbf{z})|_0 = \sum_i h_i \nabla_{\mathbf{e}} z_i|_0 = \sum_i h_i \rho_i (1, 1, 0)^T = r(1, 1, 0)^T. \quad (69)$$

The identity $J = \sqrt{(1 + 2E_{11})(1 + 2E_{22}) - (2E_{12})^2}$ gives $\nabla_{\mathbf{e}} J|_0 = (1, 1, 0)^T$. Holding r fixed when differentiating therefore yields

$$\nabla_{\mathbf{e}} [H(\mathbf{z}) - r(J - 1)]|_0 = r(1, 1, 0)^T - r(1, 1, 0)^T = \mathbf{0}. \quad (70)$$

A single scalar correction suffices because the paired features give an isotropic reference stress, even when the response away from the reference is anisotropic. The factor $J - 1$ also makes the correction zero in value at the reference. The growth terms below preserve both reference conditions.

- (iii) *Penalizing collapse and large expansion.* The term $\alpha(J - 1 - \log J)$ diverges as $J \rightarrow 0^+$, while $\beta(J - 1)^2/2$ grows quadratically as $J \rightarrow \infty$. Both have zero value at $J = 1$. Their derivatives are

$$\frac{d}{dJ} [\alpha(J - 1 - \log J)] = \alpha(1 - J^{-1}), \quad \frac{d}{dJ} \left[\frac{\beta}{2} (J - 1)^2 \right] = \beta(J - 1). \quad (71)$$

These derivatives vanish at $J = 1$, so their strain gradients also vanish by the chain rule. The quadratic term alone would remain finite at collapse, with limit $\beta/2$.

- (iv) *Penalizing distortion at bounded area.* Terms depending only on J cannot detect changes of shape that preserve area. For example, with stretch parameter $\lambda > 0$,

$$\mathbf{F} = \text{diag}(\lambda, \lambda^{-1}), \quad J = 1, \quad \varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right) = \frac{\varepsilon}{2} (\lambda^2 + \lambda^{-2} - 2). \quad (72)$$

This contribution diverges as $\lambda \rightarrow \infty$, although the area remains constant. More generally, $\text{tr } \mathbf{C} = \|\mathbf{F}\|_F^2$ supplies growth under unbounded deformation at bounded positive J . At the reference, the constant -1 sets its value to zero. Using $\mathbf{C} = \mathbf{I} + 2\mathbf{E}$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$, its stress contribution is

$$\partial_{\mathbf{E}} \left[\varepsilon \left(\frac{\text{tr } \mathbf{C}}{2} - 1 - \log J \right) \right] \Big|_0 = \varepsilon [\mathbf{I} - J^{-1} \partial_{\mathbf{E}} J|_0] = \mathbf{0}. \quad (73)$$

Zero value and zero first derivative at the reference do not imply zero second derivative: the growth terms contribute to the reference stiffness. The determinant stress correction can also change the strain tangent, since J is nonlinear in $\mathbf{e}$. The following proposition summarizes the guarantees for the complete sum.

Affine determinant stress normalization has an established precedent in Linden et al. [9]. The role of Eq. (61) here is to make that scalar normalization compatible with this anisotropic paired family. The sign of r is unrestricted. Reference normalization is separated from the independent positive barrier coefficient α and quadratic coefficient β . The equations may be read in physical energy-density units; the implementation uses a positive energy scale, which does not alter any of the following properties.

Proposition 2. *Let the features satisfy Eq. (59), let the core be either of the C^2 convex, componentwise nondecreasing constructions in Section 4.4, and let $\alpha, \beta, \varepsilon > 0$. Equation (66) is a twice continuously differentiable, objective, two-dimensional polyconvex energy on $\text{GL}^+(2)$. With $\mathbf{s} = \nabla_{\mathbf{e}} W$, it has potential structure and symmetric material stress, satisfies $W(\mathbf{I}) = 0$ and $\partial_{\mathbf{e}} W(\mathbf{0}) = \mathbf{0}$, and diverges under collapse, unbounded expansion, and unbounded distortion at bounded positive area. These are criteria (C1)–(C6). Global nonnegative energy is not asserted by this proposition.*

The proof is given in Appendix B.3. Its central composition argument is that convex features remain convex after a convex, componentwise nondecreasing core. The reference subtraction is constant, the stress correction is affine in the independent determinant variable, and the remaining analytic terms are convex in independent $(\mathbf{F}, J)$. These properties preserve polyconvexity of the complete sum; the proof also shows that the neural contribution cannot offset the prescribed growth barriers.

The proposition holds for either core and does not depend on the training error or the locations of the training samples. It is therefore a construction proof, not an inference from sampled positive tangents. Its scope is nevertheless constitutive: it does not prove uniqueness of a structural equilibrium, absence of geometric buckling, or convergence of a particular Newton algorithm. Nor does it prove that a homogenized material with an arbitrary microscopic instability can be accurately represented by a globally polyconvex law.

4.6 A separate nonnegative-energy check

Zero reference energy and stress do not, on their own, establish global nonnegative energy. This distinction is also explicit in Linden et al. [9]. We seek a lower bound $g(J)$ that depends only on the area ratio: if this bound is nonnegative for every $J > 0$, then the energy is nonnegative for every admissible deformation.

For Eq. (66), define $k_i = 1/p_i + 1/q_i$ and $\eta_i = \rho_i/k_i$. The determinant inequality $\|\mathbf{F}\mathbf{d}_i\|^2 \|\mathbf{F}\mathbf{d}_i^\perp\|^2 \geq J^2$ gives $z_i \geq k_i J^{\eta_i}$. Together with $\text{tr } \mathbf{C}/2 \geq J$ and the supporting plane of H , this yields

$$W \geq g(J) := \sum_i h_i k_i [J^{\eta_i} - 1 - \eta_i(J - 1)] + (\alpha + \varepsilon)(J - 1 - \log J) + \frac{\beta}{2}(J - 1)^2. \quad (74)$$

Since $g(1) = g'(1) = 0$, proving $g'' \geq 0$ for all $J > 0$ makes g convex with a global minimum of zero at $J = 1$. Hence $W(\mathbf{F}) \geq g(\det \mathbf{F}) \geq 0$. Appendix C describes the implemented bound. This is a saved-parameter certificate with numerical margins, not a restriction enforced at every training iterate and not a formally verified interval-arithmetic proof.

4.7 From the proof to implementation checks

With the baselines, paired features, neural cores, and complete-energy guarantees now defined, Table 2 summarizes the four learned models. Its entries refer to the implementations used here, not to every method bearing the same model name. The separate nonnegative-energy check is not included among the architectural guarantees.

Table 2: Learned constitutive models in the present comparison, with criteria ordered as in Section 4.1. “Not imposed” is not synonymous with “fails at every state”. All four models permit anisotropy without enforcing a named material symmetry group (C3).

	Regression	Free	ICNN	ICKAN
Network input	$E_{11}, E_{22}, \gamma_{12}$	$C_{11} - 1, C_{22} - 1$ $C_{12}, J - 1$	32 paired features	32 paired features
Network output	Three stresses	Scalar energy	Scalar energy core	Scalar energy core
Potential structure (C1)	Not imposed	Exact	Exact	Exact
Objectivity (C2)	Objective strain input and material stress output	Exact	Exact	Exact
Stress symmetry (C3)	Exact	Exact	Exact	Exact
Reference state (C4)	Not imposed	Exact normalization	Exact normalization	Exact normalization
Polyconvexity (C5)	Not imposed	Not imposed	By construction	By construction
Growth (C6)	No potential	Not imposed	By construction	By construction
Global $W \geq 0$	No potential	Not guaranteed	Separate saved-weight bound	Separate saved-weight bound

The analytical guarantees and their numerical verification have distinct purposes. First, the feature inequalities, positive scales, core weight signs, and analytic normalization identify the admissible model family. Second, the nonnegative-energy bound is evaluated for each selected parameter set, since this property is not guaranteed by the architecture alone. Failure of that sufficient bound would be inconclusive, not proof of a negative energy.

Finally, derivative and constitutive tests check that the implemented model represents the stated equations. These include reference energy and stress, first and second energy derivatives, the analytically assembled stress against energy differentiation, spline derivatives at their joins, and the physical-unit conversion of stress and tangent. Closed-cycle work checks potential consistency, while sampled rank-one curvatures test a consequence of polyconvexity using Eq. (88), including its stress-dependent term. No finite set of positive curvature samples is a substitute for Proposition 2. Likewise, a negative value in an implementation of this construction would require investigation of the implementation, admissible domain, or numerical differentiation before being interpreted as a constitutive result.

Section 5 reports the selected-model checks and predictive errors separately. This separation allows a model to be accurate on the data but lack a constitutive guarantee, or to possess that guarantee while retaining a nonzero approximation error.

5 Constitutive assessment on periodic microstructures

This assessment separates constitutive approximation accuracy from mechanical guarantees and from the structural deployment considered in Section 6. Material A is the single-cavity periodic cell for which results are available. Material B denotes a planned multicavity cell; it is not yet a validated second material.

Draft task—evidence pending. This section is the working skeleton for a two-material assessment. All numerical results currently reported concern material A. Geometry verification, data generation and model evaluation for B, as well as the controlled fixed/learned-feature comparison, remain pending. The abstract, introduction and conclusions still describe the available evidence and will be reconciled after that campaign.

5.1 Microstructures and reference-model verification

5.1.1 Material A: the available single-cavity reference

The square periodic cell contains a centered elliptical void with area fraction 0.20, aspect ratio 2, and major-axis angle 30° relative to the cell axes; see Fig. 4. This orientation avoids the mirror-aligned configurations of the square cell. Centrosymmetry remains, but does not mean that the strain energy is an even function of strain.

The solid is compressible Neo-Hookean in plane strain:

$$\psi_\mu = \frac{\mu}{2}(\text{tr } \mathbf{C}_\mu - 2) - \mu \log J_\mu + \frac{\lambda}{2}(\log J_\mu)^2, \quad (75)$$

with Young’s modulus 1.628 GPa and Poisson ratio 0.4. Here μ and λ are the usual Lamé constants. This is the specific matrix law used by the reference solver, not a generic assertion about every Neo-Hookean variant. The production mesh contains 1546 quadratic triangles, 3240 nodes, and 4638 quadrature points. Periodicity and null-space constraints leave 6320 independent microscopic displacement coordinates. Cell averages use the complete reference cell measure.

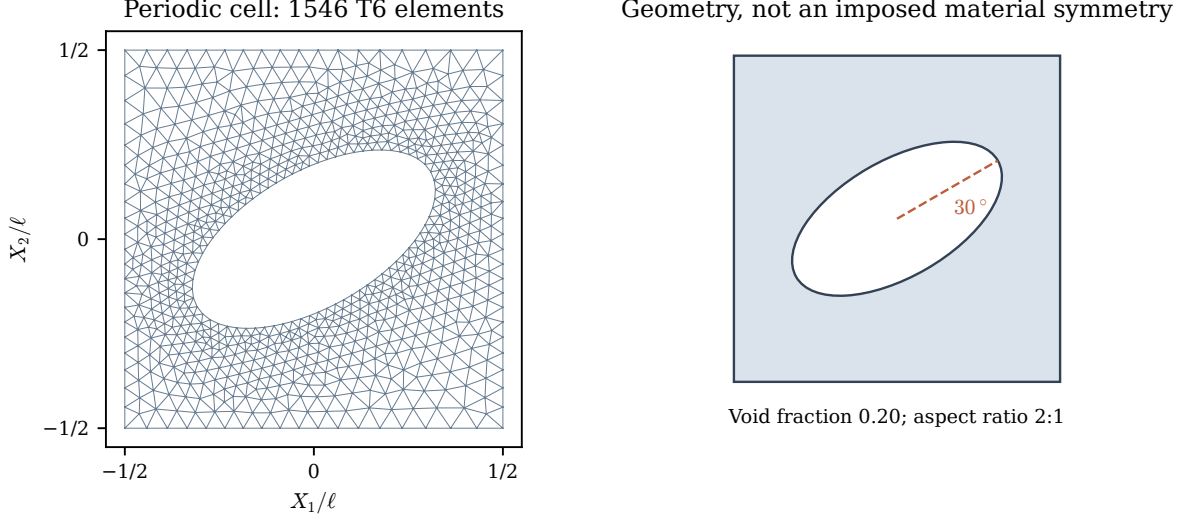


Figure 4: Microstructure and production discretization. The left panel shows corner-edge traces of the quadratic mesh; the right panel specifies the void geometry. The cell side is denoted by ℓ . No material reflection symmetry is imposed on the learned paired features.

The production mesh was selected using a local stress-field criterion along an investigated uniaxial path, rather than only a homogenized average. At $E_{11} = 0.20$, its maximum microscopic first Piola stress norm differs by approximately 0.486% from a 3095-element reference, whereas the average axial stress differs by approximately 0.001%. This is a mesh check at a specified state, not a uniform error bound over all multiaxial data.

5.1.2 Material B: proposed multicavity reference

Draft task—evidence pending. Define a genuinely different periodic cell with four elliptical cavities of different sizes and orientations, initially retaining the matrix law, plane-strain setting and total porosity of material A. Finalize positions, semiaxes and angles only after checking periodic geometry, solid connectivity and mesh quality. A FOM pilot must establish reliable multiaxial reference responses and a justified data domain; no geometry, mesh or response for B is reported as verified here.

5.2 Data domains and learning protocol

Material A: available data and selected models. The fitting domain is informed by a preliminary macroscopic deformation study. The final training box, shared by the constitutive and reduced models, has the following rounded bounds:

$$\mathbf{e}_{\min} = (0, -0.09865, -0.16807)^T, \quad \mathbf{e}_{\max} = (0.1955, 0.01635, 0.10793)^T. \quad (76)$$

The new constitutive features do not change this box. For the selected polyconvex network runs, 4208 of the 4950 available training states are used for fitting and 742 for validation, with split seed 5. The independent test set contains 400 states. The additional out-of-domain probe, its failed reference solves and the common finite-label mask are documented in Supplementary Section S1.

The selected feature-learning runs use full-batch Adam for 2600 steps followed by up to 400 L-BFGS iterations. The normalized training objective combines stress, energy, and reference-tangent relative mean-square errors with weights 1, 0.2, and 0.02, respectively. The tangent target is obtained from a separate coarse

periodic reference calculation. Candidates are selected by validation stress error; test and probe labels are not used in that second-stage selection. Both selected runs use initialization seed 16, with 32 features and hidden widths 24/24 for ICNN and 12/12 for ICKAN. Earlier baselines have different feature families, objectives, and training budgets. Comparisons to them diagnose practical approximability, not compute-matched superiority of one network architecture.

For the selected integrated-hat ICKAN, every connection uses the same six fixed spline bases, with knot centers κ_k uniformly spaced from -1.2 to 1.2 , giving $h = 0.48$ in the scalar input coordinates of each spline layer.

For both selected models, the input center $z_i^{\text{cen}} = z_i^0$ is recomputed from the current stretch powers during training. Each scale s_i is set once before optimization to the maximum absolute deviation of feature z_i from its reference value over the fitting set, evaluated with the initialized feature parameters and bounded below by 10^{-4} ; it is then held fixed.

Draft task—evidence pending. Add a compact data-coverage figure for A and, once available, B. Identify the sampling pattern, fitting/validation split and independent test states; declare any projections or subsampling. Freeze the domain, sample counts, independent loading paths, optimization budgets and initialization seeds for B before final evaluation. Its domain must not be inherited automatically from A’s coupon-derived box.

5.3 Constitutive response and predictive accuracy

For material A, the selected ICNN and ICKAN models attain held-out aggregate stress errors of approximately 0.65%; Table 3 reports the independently audited selected checkpoints. Free has a corresponding test error of 1.2643%. This comparison concerns the archived training protocols, not a matched-budget architecture study. Outside the training domain, Free has a lower aggregate probe error than either polyconvex model; Supplementary Section S1 preserves those results. Architectural admissibility does not guarantee extrapolation accuracy.

Table 3: Material A: independent-test constitutive errors of the selected polyconvex models, in percent.

Model	e_S^{test}	e_W^{test}	Worst test
ICNN	0.6491	0.2169	5.53
ICKAN	0.6551	0.2148	5.78

All values are percentages. Stress and energy errors are aggregate array norms; the last column is the maximum per-state stress error. Material A: 400 independent test states.

Draft task—evidence pending. Add FOM/model stress curves on predeclared axial, shear and combined loading paths, followed by the corresponding independent-test results for B. Inventory existing evaluations before requesting new FOM solves; do not connect unrelated scattered states as though they were a loading path. Report componentwise behavior and variability where needed to interpret aggregate errors.

5.4 Contribution of learned paired features

Draft task—evidence pending. Run a controlled comparison of fixed and learned admissible paired features, preferably for both A and B. Keep the core architecture, feature count, data, objective and optimization budget comparable; declare the fixed configuration and multiple initialization seeds before inspecting test errors. Retain model-specific reference normalization and growth terms. This comparison is not supplied by the archived baselines and has no results yet.

5.5 Mechanical consistency and scope

The following numerical checks concern the available material-A checkpoints. They complement, rather than replace, the construction guarantees in Section 4.

The archived independent audit records successful energy first- and second-derivative checks and saved-weight nonnegative-energy sufficient bounds for both selected models. The conservative finite-interval margins are 17.25 and 16.82 in normalized energy units. Random admissible-deformation tests support implementation consistency, but the polyconvexity argument is analytical. Neither aggregate error nor a successful bound on energy is a uniform bound on stress prediction error.

5.5.1 Closed-cycle tests of constitutive integrability

For a differentiable hyperelastic potential, any closed strain cycle in its domain satisfies

$$\mathcal{W}_{\text{cyc}} = \oint \mathbf{s} \cdot d\mathbf{e} = 0. \quad (77)$$

This test isolates integrability and does not require a structural Newton failure. The importance of conservative strain–stress mappings is discussed explicitly by As’ad et al. [7]. We evaluate a counterclockwise rectangular cycle in (E_{11}, E_{22}) , with half-width 0.001, centered at

$$\mathbf{e}_c = (0.0051260033, -0.0945507613, -0.1283720602)^T. \quad (78)$$

The engineering shear component is held fixed. The rectangle lies inside Eq. (76). Its center was chosen after inspecting regression tangent asymmetry at held-out states; it is a deliberately selected mechanical witness, not an unbiased estimate of the frequency of violations. The models remain frozen during this audit.

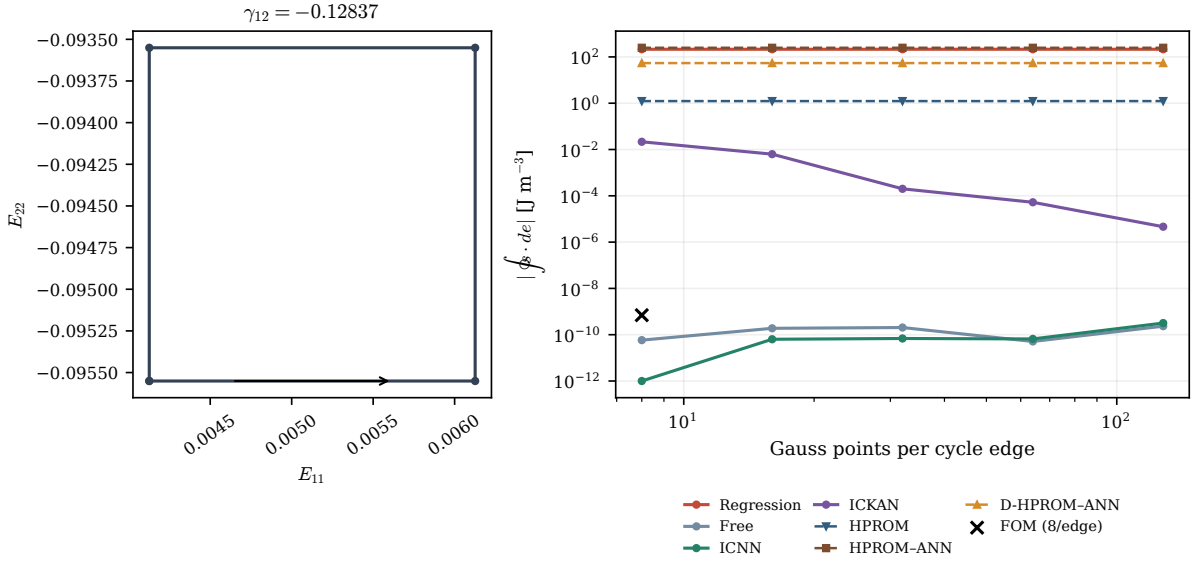


Figure 5: Closed-cycle geometry and quadrature refinement of net stress work. Regression and all three deployed HPROM variants retain nonzero work; signed values are reported in Table 4. Free and ICNN cancel to numerical precision; an exact numerical zero is plotted at 10^{-12} J m $^{-3}$ solely for logarithmic display. The conservative C^2 spline energy requires quadrature refinement when crossing knots; its residual work decreases towards zero. The FOM marker is a separately evaluated cycle at eight Gauss points per edge.

Table 4 retains the signed work and its quadrature dependence. Reversing the cycle reverses the regression work, which is incompatible with interpreting that map as an elastic potential. The result establishes a constitutive inconsistency, not by itself a quantitatively important error for every macroscopic loading history. The Free model passes this test because it is energy-based; energy-cycle consistency therefore cannot distinguish Free from the polyconvex models.

The same cycle tests all three frozen HPROM variants. The affine HPROM and iterative HPROM–ANN re-solve reduced equilibrium at each quadrature point; D–HPROM–ANN directly reconstructs the state and evaluates the stress. Their net work converges to -1.23416 , $+247.774$, and $+54.0815$ J m $^{-3}$, respectively. Independent reverse traversals recover the opposite signs, with forward-plus-reverse work below 6×10^{-7} J m $^{-3}$ in magnitude. Tightening the microscopic displacement-increment tolerance from 10^{-10} to 10^{-12} changes the 128-point-per-edge work by less than 4×10^{-10} J m $^{-3}$ for HPROM and 3×10^{-7} J m $^{-3}$ for HPROM–ANN. No equilibrium-tolerance control applies to the direct variant. The nonzero values therefore persist under the tested quadrature and solver refinements.

These results concern the deployed constitutive output maps, not a general failure of hyperreduction to preserve energy. In particular, the affine fixed-weight HPROM has an exact reduced equilibrium Jacobian, but its independently fitted stress cubature is not constructed as the derivative of the reduced equilibrium energy. A potential structure for the reduced residual alone therefore does not certify integrability of this effective stress output. The nonlinear variants add decoder and state-dependent cubature approximations, as discussed in Section 3. The cycle detects the resulting output inconsistency but does not isolate its dominant

cause. Although the affine HPROM defect is much smaller on this path, neither its size nor the ranking on one cycle establishes the error on arbitrary loading histories.

This audit uses the pre-existing cycle, not a path selected to maximize a reduced-model defect. At 128 points per edge, the instrumented HPROM-ANN run records 2416 modified-Newton iterations including 202 in the initial continuation from zero; subsequent quadrature queries require four to five iterations. These counts describe this diagnostic only and are not substituted for microscopic iteration counts in the archived structural timing runs.

Table 4: Closed-cycle work under quadrature refinement.

Model	8	16	32	64	128
Regression	-213.922	-213.922	-213.922	-213.922	-213.922
Free	-5.82077e-11	1.89175e-10	2.04182e-10	-5.10454e-11	2.33996e-10
ICNN	0	-6.36646e-11	6.86668e-11	-6.5711e-11	3.18607e-10
ICKAN	-0.0215596	0.00629948	-0.000200597	-5.25639e-05	4.62179e-06
HPROM	-1.23416	-1.23416	-1.23416	-1.23416	-1.23416
HPROM-ANN	247.774	247.774	247.774	247.774	247.774
D-HPROM-ANN	54.0815	54.0815	54.0815	54.0815	54.0815

Signed cycle work in J m^{-3} ; column headings give Gauss points per edge. FOM at 8 points per edge: $-6.98492\text{e-}10 \text{ J m}^{-3}$. The spline quadrature error decreases with refinement; it is not physical dissipation. Reversal and equilibrium-tolerance controls are discussed in the text.

5.5.2 Curvature checks and limits of the evidence

For the energy-based models, sampled rank-one curvatures are evaluated with the full stress-dependent expression in Eq. (88). The detailed sampling audit and the outward search for a negative Free-model curvature are retained in Supplementary Section S2. That negative candidate is confirmed for the learned energy but lacks a converged FOM reference; it does not establish an artificial instability relative to a verified stable homogenized response. Positive samples for ICNN and ICKAN check implementation, whereas their polyconvexity guarantee is analytical.

Draft task—evidence pending. Repeat the reference-state, derivative, cycle and saved-parameter nonnegative-energy checks for the selected material-B models. Report bounds that fail or remain inconclusive. Existing A-checkpoint audits cannot certify B, and a second material trained separately does not demonstrate transfer between geometries.

6 Structural deployment for material A: accuracy and online cost

6.1 Purpose, boundary conditions, and coverage

This example asks how the constitutive approximations affect a complete nonlinear structural calculation. The material is the single-cavity cell A defined in Section 5.1.1; the new evidence concerns error propagation and online cost, not a second microstructure. The dog-bone profile has overall length 165 mm, grip width 19 mm, narrow-section width 13 mm, narrow-section length 57 mm, and fillet radius 76 mm. It is a plane-strain computational benchmark using an ASTM-inspired in-plane profile, not a simulation certified to reproduce an ASTM tensile test.

The two end faces receive equal-and-opposite axial resultants of 100 kN per end, applied in 20 equal increments. Three point constraints remove rigid modes: the axial and transverse displacement at the left mid-edge node, and transverse displacement at the right mid-edge node. Entire end faces are not clamped. Consequently, a small skew of the deformed ends is not ruled out by the boundary conditions. The anisotropic response can couple axial extension to shear.

The macro mesh contains 690 quadratic triangles, 1501 nodes, and 2070 integration points. The reference extrusion thickness used to convert the two-dimensional stress integrals into the stated force resultants is 50 mm. This force convention is part of the plane-strain benchmark, not an ASTM specimen-thickness specification. The physical cell size is declared as 1.3 mm, corresponding to ten cells across the narrow width. This provides a scale-separation assumption, not a direct numerical validation of homogenization error at that ratio. All saved converged integration-point strains for every reported model remain inside the unchanged training box. This check does not automatically cover unsaved intermediate Newton iterates or certify another load case.

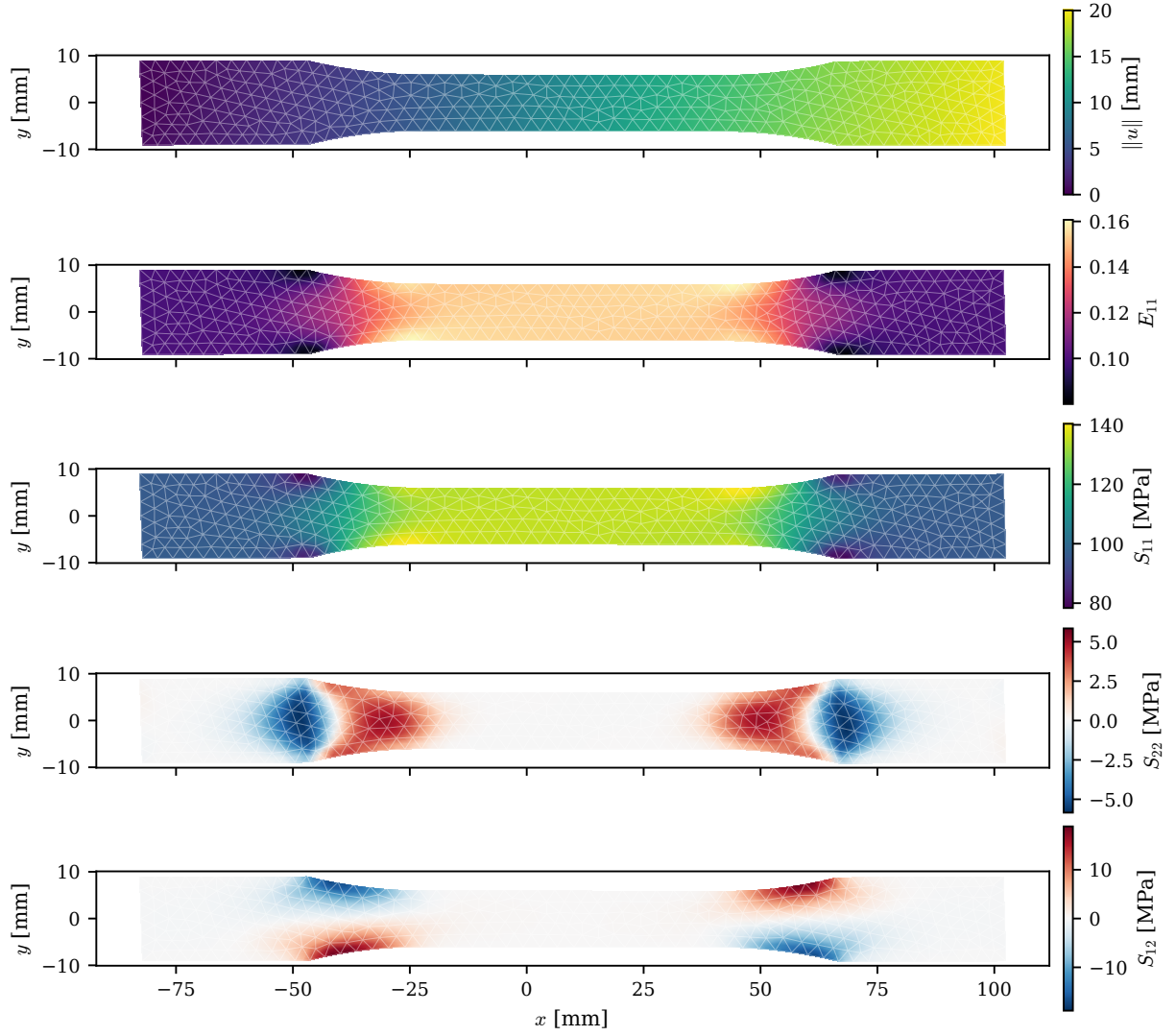


Figure 6: Nested FOM–FE² reference at 100 kN per end. Fields are displayed on the actual deformed mesh, without displacement amplification. Gauss-point strain and stress values are first averaged within each element and then patch-averaged to nodes for visualization only. These recovered contours are not the raw fields used in error norms and can smooth local extrema. Stresses are second Piola components. A three-dimensional von Mises stress is not reconstructed because the saved output does not include the necessary out-of-plane Cauchy stress.

6.2 Reduced-state representation for material A

The 4950 displacement snapshots provide the basis of the reduced microscopic models as well as the effective constitutive labels. Figure 7 separates POD compression from the strain-informed coordinate fit. The relative snapshot truncation norm for the retained 39-dimensional space is 9.94×10^{-7} . After its orthogonal rotation into primary and secondary spaces, $n = 3$ and $\bar{n} = 36$ in the notation of Section 3.

The fitted relation between the deployed primary coordinates and macroscopic strain has an aggregate relative array discrepancy of 3.60×10^{-5} on these snapshots; its maximum absolute component discrepancy is 2.55×10^{-5} . These quantities explain why macroscopic strain is a useful primary-coordinate predictor for this material. They do not imply exact equality at unsampled states, nor do they measure the closure error or the equilibrium residual. In particular, the direct substitution in D-HPROM-ANN is an additional approximation, even when the plot appears visually indistinguishable from the identity.

The secondary-coordinate closure is a 3–64–64–36 feed-forward network with two tanh hidden layers and a linear output. Its training script uses input standardization, the raw secondary-coordinate loss in Eq. (23), and a 90/10 internal fit/validation split with seed 20260903. Full-batch Adam has initial learning rate 2×10^{-3} , a validation-driven reduction schedule, and a maximum budget of 20000 epochs. The archived best relative validation mean-square error is 3.75×10^{-7} . This is an internal closure metric after a POD built from all 4950 snapshots, not an independent material-generalization estimate. Independent effective-stress and structural tests provide the latter assessment.

Each deployed adaptive cubature field has architecture 3–128–128–128–10, GELU hidden activations, and the scaled softmax output of Eq. (39). The residual and stress checkpoints use distinct ten-element supports. Their larger neural cores illustrate why the online cost cannot be inferred solely from the three equilibrium coordinates or from the element count. These reduction-training choices differ from those of the constitutive networks; the common material and data domain do not imply matched offline training budgets.

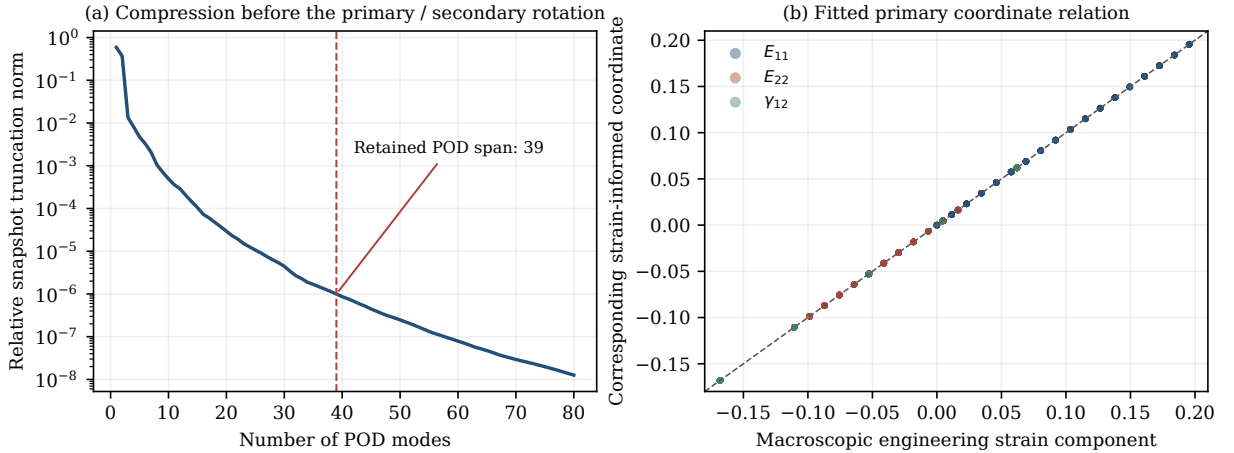


Figure 7: Diagnostics from the deployed 39-mode basis artifact. Left: relative POD snapshot truncation norm, before the primary/secondary rotation. Right: strain-informed primary coordinates versus the corresponding macroscopic strains; every tenth snapshot is plotted, whereas the reported discrepancies use all 4950 states. Neither diagnostic measures structural accuracy or proves validity outside the sampled domain.

6.3 Online models and implementation verification

The linear HPROM uses $n_{\text{tra}} = 39$ generalized coordinates, a residual rule supported on 135 elements, and a stress rule on 73 elements; their union contains 183 elements. The integrand-SVD tolerances are $\varepsilon_{\text{SVD}}^r = 10^{-3}$ and $\varepsilon_{\text{SVD}}^s = 10^{-5}$, retaining 134 and 72 integrand modes, respectively. The greedy integration tolerance in Eq. (36) is $\varepsilon_{\text{ECM}} = 10^{-6}$ for both rules, including the multiplier-sum constraint.

The iterative HPROM-ANN uses $n = 3$ primary and $\bar{n} = 36$ secondary coordinates, with ten residual and ten stress elements and a union of 19. Initial fixed-ECM candidates use $\varepsilon_{\text{SVD}} = 10^{-4}$ and $\varepsilon_{\text{ECM}} = 10^{-6}$; adaptive pruning targets ten elements per rule. Thus this final count is a prescribed pruning target assessed through integration and response errors, not a consequence of the greedy tolerance alone. The support-generation driver uses up to 500 sampled states and tests up to 20 removal candidates per elimination. Its positivity-enforcing phase is active from the first elimination; the available driver does not enable or supply the global graph coupling. The residual weight field is subsequently refitted on the same pruned support

with a longer training budget. Historical pruning diagnostics are not stored in the deployed checkpoints, so their filenames alone are not evidence of a graph-regularized run.

The direct model evaluates stress on ten elements and solves no microscopic equilibrium equations. All these counts refer to reduced computational meshes, not full-mesh evaluations followed by masking. Figure 8 shows their spatial distribution and the overlap of residual and output rules.

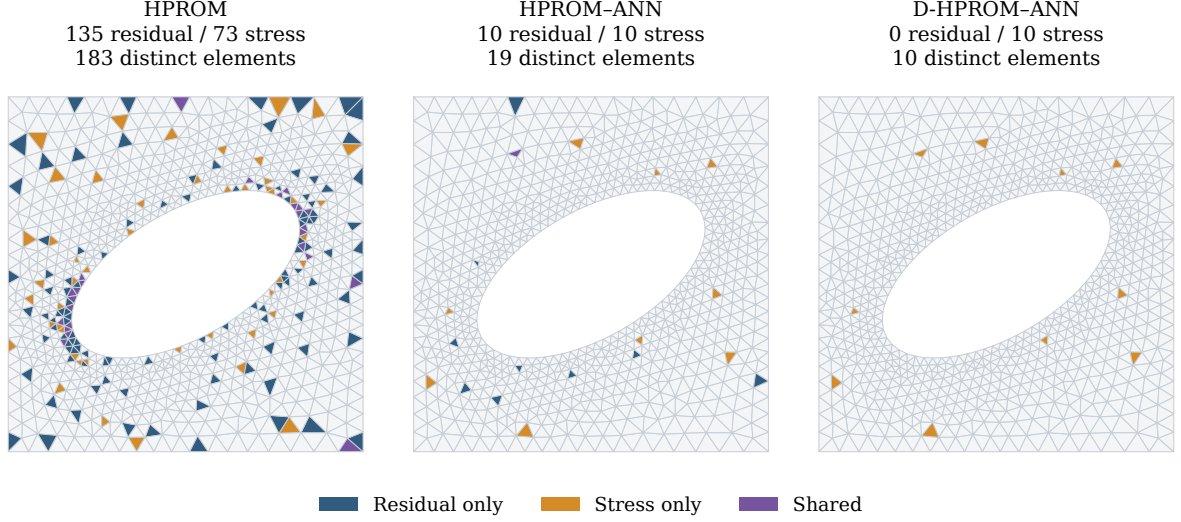


Figure 8: Element supports of the deployed hyperreduced microscopic models. The pale full-cell mesh provides location context only; discarded elements are not assembled online. Colors distinguish residual-only, stress-only, and shared elements. Supports are read from the same index arrays used to construct the reduced MDPA meshes.

For the optimized linear and direct implementations, the maximum normalized stress differences from the original implementations over the verification states are 4.71×10^{-14} and 4.20×10^{-13} . Corresponding tangent differences are 2.68×10^{-10} and 7.25×10^{-10} . Independent finite-difference checks give discrepancies of approximately 3.51×10^{-9} and 2.38×10^{-9} . Such comparisons check that optimization preserves the deployed model; they do not establish that either model exactly reproduces full microscopic equilibrium.

6.4 Error and timing definitions

For a saved nodal or integration-point array A , the reported relative error is

$$e_A = 100 \frac{\|\text{vec}(A - A_0)\|_2}{\|\text{vec}(A_0)\|_2}, \quad (79)$$

where A_0 is the nested reference on the same meshes. These are unweighted discrete array norms. The strain array stores engineering shear, whereas the stress array stores tensor shear. They are neither quadrature-weighted continuum L^2 norms nor invariant tensor Frobenius norms. The metric is kept identical across models; this does not make the three different field errors interchangeable.

Online wall time includes the complete macroscopic solve, microscopic or surrogate stress/tangent evaluations, communication, and lazy worker construction within the timed solve. Parent setup, offline data generation, training, reduced-mesh construction outside the timed region, and result export are excluded. The FOM uses 20 worker processes with one numerical-library thread per worker. The three optimized microscopic surrogates use the same process configuration and three sequential fresh complete runs. PANN evaluations are batched in one process and use the separately investigated thread settings shown in Table 5.

The macroscopic solve stops when the free-degree-of-freedom residual norm falls below 10^{-5} N or its ratio to the initial increment residual falls below 10^{-7} . The periodic FOM uses a relative microscopic displacement-increment tolerance of 10^{-10} , with continuation when needed. The archived FOM and selected PANN times each have one repetition in this comparison. Thus their ratios are observed online speedups under the stated implementations, not matched-thread efficiency estimates or repeated-run confidence intervals. Table 6 exposes the available reduced-model variability rather than selecting its fastest repetition. A publication-final timing study should repeat every tier under a common resource-budget protocol and retain all repetitions.

The affine HPROM uses its exact reduced residual Jacobian, whereas HPROM-ANN uses the projected-stiffness modified-Newton matrix described in Section 3.4.2; both use a microscopic increment tolerance

of 10^{-10} . Their effective constitutive tangents nevertheless account for the equilibrium sensitivity of their respective full residuals. The archived structural runs do not retain microscopic iteration totals. The reported speedup therefore compares these implemented workflows, including their different iteration matrices and derivative costs; it does not isolate a nonlinear-representation gain or quantify the iteration penalty of modified Newton.

Table 5: Complete coupon simulations: final-field errors against the nested FOM and archived online wall times.

Model	CPU	n_{run}	t [s]	t_0/t	e_u [%]	e_E [%]	e_S [%]
FOM	20p	1	4268.679	1.0	0.00000	0.00000	0.00000
HPROM	20p	3	67.348	63.4	0.00565	0.01653	0.00099
HPROM-ANN	20p	3	30.143	141.6	0.05200	0.14968	0.03727
D-HPROM-ANN	20p	3	3.788	1126.9	0.02324	0.04851	0.00967
ICNN	4t	1	1.734	2461.3	0.42468	0.77028	0.08880
ICKAN	8t	1	3.669	1163.5	0.38657	0.78932	0.07800
Free	12t	1	1.356	3146.9	0.92988	1.49464	0.13492
Regression	16t	1	0.946	4510.4	0.60620	0.81218	0.07378

p: worker processes, one numerical-library thread each; t: PyTorch threads. For $n_{\text{run}} = 3$, time is the median; for $n_{\text{run}} = 1$, it is the archived single run. These are online wall times, not offline-inclusive or matched-thread speedups.

Table 6: Sequential optimized-run repetitions used for median HPROM timings.

Model	t_1 [s]	t_2 [s]	t_3 [s]
HPROM	63.354	67.348	68.953
HPROM-ANN	27.947	34.628	30.143
D-HPROM-ANN	3.788	3.758	3.969

6.5 Accuracy–cost trade-off and Newton behavior

Figure 9 and Table 5 show distinct operating points rather than a single universally best model. The linear HPROM gives the smallest displacement error, approximately 0.00565%, at a median online time of 67.35 s. The iterative HPROM-ANN reduces this time to 30.14 s with a displacement error of approximately 0.0520%. The optimized iterative nonlinear model is therefore about 2.23 times faster than the *optimized* linear model; comparison against the old linear implementation would overstate this gain. The direct closure reduces the time to 3.79 s and has a displacement error of approximately 0.0232% on this path. Its lower error than the iterative nonlinear model is an observed outcome, not a consequence of a general error ordering.

The ICNN and ICKAN times are 1.73 s and 3.67 s, with displacement errors of approximately 0.425% and 0.387%. Their finite-strain structural accuracy is therefore consistent with a useful constitutive approximation, although it is poorer than the reduced microscopic models on this benchmark. Direct stress regression is fastest, but its successful coupon solution does not remove the integrability defect exposed by Section 5.5.1. Free also converges and remains a relevant baseline; its absence of a global convexity constraint is not equivalent to a failure on this load case.

All models require four macroscopic iterations per increment in the archived runs. Consequently, this example provides no empirical evidence that polyconvexity reduced Newton iterations relative to Free or Regression. It establishes that the selected constrained models can be used successfully with the common structural solver, while maintaining their constitutive guarantees. A separate challenge problem would be needed to establish a solver-robustness advantage, and its reference solution would need to be validated independently.

7 Conclusions

Paired directional features with constrained trainable powers and determinant exponents provide a practical polyconvex constitutive construction for the anisotropic periodic material studied here. Their isotropic reference derivatives allow exact stress normalization through an affine determinant correction without tying the normalization coefficient to the volumetric barrier stiffness. The proof combines feature convexity in independent minors with core convexity and monotonicity, and includes every correction in the complete energy. Global nonnegative energy is supported separately by a sufficient bound evaluated for the selected

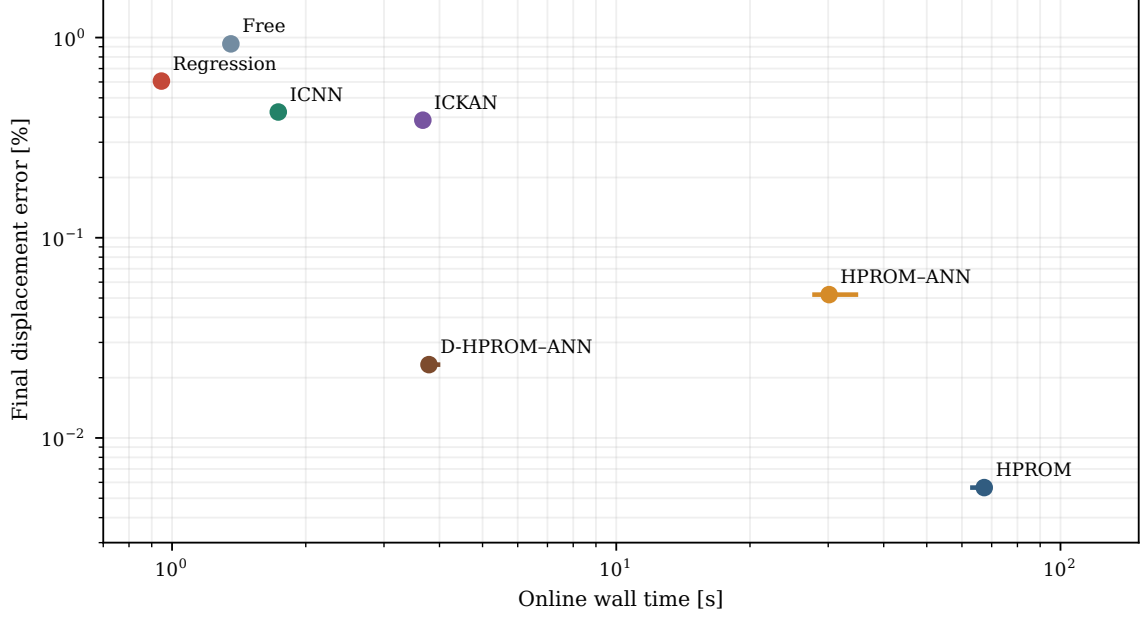


Figure 9: Online cost versus final displacement error on the common coupon problem. Horizontal segments show the observed range of three optimized HPROM runs; unbarred points are archived single PANN runs. Process and thread counts differ as detailed in Table 5. The plot does not include offline costs.

parameters. Both a smooth ICNN and an integrated-hat spline-ICKAN achieve approximately 0.65% held-out aggregate stress error and below 0.43% displacement error in the nominal coupon simulation.

Optimized reduced microscopic models occupy a different accuracy–cost range. They are more accurate on this coupon, while direct learned constitutive evaluation is generally faster. A closed-cycle test exposes nonconservativity in the regression baseline and all three deployed HPROM variants, despite their successful structural solutions. In contrast, the available Free-model curvature witness is not yet supported by a converged FOM reference and is reported only within that limited scope. The study supports the usefulness of the proposed constrained feature family for the present problem, not unrestricted extrapolation or a universal ranking of constitutive and reduced-order models.

A Rank-one paths and finite-strain energy curvature

This appendix supplies the algebra behind the two finite-strain curvature statements in criterion (C5).

Determinant along a rank-one path. For arbitrary $\mathbf{F}, \mathbf{H} \in \mathbb{R}^{2 \times 2}$, expansion of the two products in the determinant gives

$$\begin{aligned} \det(\mathbf{F} + t\mathbf{H}) &= (F_{11} + tH_{11})(F_{22} + tH_{22}) - (F_{12} + tH_{12})(F_{21} + tH_{21}) \\ &= \det \mathbf{F} + t(F_{22}H_{11} + F_{11}H_{22} - F_{21}H_{12} - F_{12}H_{21}) + t^2 \det \mathbf{H}. \end{aligned} \quad (80)$$

The coefficient of t is $\text{cof } \mathbf{F} : \mathbf{H}$, since

$$\text{cof } \mathbf{F} = \begin{pmatrix} F_{22} & -F_{21} \\ -F_{12} & F_{11} \end{pmatrix}, \quad \mathbf{A} : \mathbf{B} = \sum_{i,j} A_{ij} B_{ij}. \quad (81)$$

For $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, its entries are $H_{ij} = a_i b_j$, so

$$\det \mathbf{H} = a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1 = a_1 a_2 (b_1 b_2 - b_2 b_1) = 0. \quad (82)$$

The determinant is therefore affine along a rank-one path:

$$\det(\mathbf{F} + t\mathbf{a} \otimes \mathbf{b}) = \det \mathbf{F} + t \text{cof } \mathbf{F} : (\mathbf{a} \otimes \mathbf{b}). \quad (83)$$

If $\mathbf{F}_B = \mathbf{F}_A + \mathbf{H}$ and $\mathbf{F}_s = (1-s)\mathbf{F}_A + s\mathbf{F}_B$, with $\mathbf{H} = \mathbf{a} \otimes \mathbf{b}$, then

$$(\mathbf{F}_s, \det \mathbf{F}_s) = (1-s)(\mathbf{F}_A, \det \mathbf{F}_A) + s(\mathbf{F}_B, \det \mathbf{F}_B). \quad (84)$$

For $0 \leq s \leq 1$ and positive endpoint determinants, the whole segment remains admissible. Convexity of $\mathcal{P}$ therefore yields

$$W(\mathbf{F}_s) \leq (1-s)W(\mathbf{F}_A) + sW(\mathbf{F}_B). \quad (85)$$

For a twice differentiable energy, the restriction to this line consequently has nonnegative second derivative, as required by Eq. (51).

Strain curvature and the chain rule. Let $\mathbf{H} \in \mathbb{R}^{2 \times 2}$ be an arbitrary perturbation and let t parametrize $\mathbf{F}(t) = \mathbf{F} + t\mathbf{H}$. Dots denote derivatives with respect to t , and $\text{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$ denotes the symmetric part of a matrix. Direct multiplication gives

$$\begin{aligned} \mathbf{E}(t) &= \frac{1}{2} [\mathbf{F}^T \mathbf{F} - \mathbf{I} + t(\mathbf{F}^T \mathbf{H} + \mathbf{H}^T \mathbf{F}) + t^2 \mathbf{H}^T \mathbf{H}] \\ &= \mathbf{E}(0) + t \text{sym}(\mathbf{F}^T \mathbf{H}) + \frac{1}{2} t^2 \mathbf{H}^T \mathbf{H}. \end{aligned} \quad (86)$$

Thus $\dot{\mathbf{E}}(0) = \text{sym}(\mathbf{F}^T \mathbf{H})$ and $\ddot{\mathbf{E}}(0) = \mathbf{H}^T \mathbf{H}$. Using the engineering-strain convention of Eq. (2), the chain and product rules give

$$\begin{aligned} \frac{dW}{dt} &= \mathbf{s} \cdot \dot{\mathbf{e}}, \\ \frac{d^2 W}{dt^2} &= \sum_{i,j} \frac{\partial s_i}{\partial e_j} \dot{e}_j \dot{e}_i + \sum_i s_i \ddot{e}_i = \dot{\mathbf{e}}^T \mathbf{D} \dot{\mathbf{e}} + \mathbf{S} : \ddot{\mathbf{E}}. \end{aligned} \quad (87)$$

Substituting $\ddot{\mathbf{E}}(0)$ gives

$$\left. \frac{d^2 W(\mathbf{F} + t\mathbf{H})}{dt^2} \right|_{t=0} = \dot{\mathbf{e}}(0)^T \mathbf{D} \dot{\mathbf{e}}(0) + \mathbf{S} : (\mathbf{H}^T \mathbf{H}). \quad (88)$$

Here $\mathbf{D} = \partial \mathbf{s} / \partial \mathbf{e}$ is the strain tangent and $\mathbf{S}$ is the second Piola–Kirchhoff stress, both evaluated at $t = 0$. The first term accounts for the change of stress with strain; the second accounts for the curvature of the strain path, weighted by the current stress. Even if $\mathbf{D}$ is positive semidefinite, the stress term can be negative in compression.

An analytical counterexample. Consider $W(\mathbf{E}) = \mu \mathbf{E} : \mathbf{E}$, where $\mu > 0$ is a constant with units of energy density. One has $\mathbf{S} = 2\mu \mathbf{E}$. Since

$$W(\mathbf{e}) = \mu(e_1^2 + e_2^2 + e_3^2/2), \quad \mathbf{D} = \text{diag}(2\mu, 2\mu, \mu), \quad (89)$$

the energy is strictly convex in engineering strain. At $\mathbf{F} = \lambda \mathbf{I}$, where $\lambda > 0$ is the common in-plane stretch, choose the rank-one perturbation $\mathbf{H} = \text{diag}(1, 0)$. Then

$$\dot{\mathbf{e}} = (\lambda, 0, 0)^T, \quad \mathbf{S} = \mu(\lambda^2 - 1)\mathbf{I}, \quad \mathbf{H}^T \mathbf{H} = \text{diag}(1, 0). \quad (90)$$

Hence the strain-tangent term is $2\mu\lambda^2$, and the stress term is $\mu(\lambda^2 - 1)$, giving

$$\left. \frac{d^2 W(\lambda \mathbf{I} + t\mathbf{H})}{dt^2} \right|_{t=0} = 2\mu\lambda^2 + \mu(\lambda^2 - 1) = \mu(3\lambda^2 - 1). \quad (91)$$

At $\lambda = 1/2$, the curvature is $-\mu/4$, violating Eq. (51) despite convexity in $\mathbf{E}$. This is an analytical counterexample, not a sampled observation about the RVE or a trained model. Compression alone does not force negative curvature: at $\lambda = 3/4$, the same expression is $11\mu/16 > 0$.

Equivalently, direct substitution gives

$$W(\lambda \mathbf{I} + t\mathbf{H}) = \frac{\mu}{4} [(\lambda + t)^2 - 1]^2 + \frac{\mu}{4} (\lambda^2 - 1)^2, \quad (92)$$

whose second derivative at zero is $\mu(3\lambda^2 - 1)$. The path is admissible near zero because $\det(\lambda \mathbf{I} + t\mathbf{H}) = \lambda(\lambda + t) > 0$ for $t > -\lambda$.

B Proofs for the paired features and complete energy

We first establish joint convexity of the directional building block, then prove the paired-feature properties in Proposition 1. Finally, we establish the complete-energy guarantees in Proposition 2.

B.1 Joint convexity of the directional building block

Consider $f(\mathbf{x}, J) = \|\mathbf{x}\|^a J^{-b}$ with $J > 0$, $a \geq 1$ and $0 \leq b \leq a - 1$. For $t = \|\mathbf{x}\| > 0$, the Hessian block in the radial and determinant variables is

$$\begin{pmatrix} a(a-1)t^{a-2}J^{-b} & -abt^{a-1}J^{-b-1} \\ -abt^{a-1}J^{-b-1} & b(b+1)t^aJ^{-b-2} \end{pmatrix}. \quad (93)$$

Its diagonal entries are nonnegative and its determinant is

$$ab(a-b-1)t^{2a-2}J^{-2b-2} \geq 0. \quad (94)$$

The transverse Hessian eigenvalues are $at^{a-2}J^{-b} \geq 0$. Continuous extension gives convexity at $\mathbf{x} = 0$. For the constitutive evaluation on $\text{GL}^+(2)$, a nonzero reference direction cannot satisfy $\mathbf{F}\mathbf{d} = 0$, so this point does not introduce an interior singularity of the directional norm. Taking $a = 2p$ or $2q$ establishes Eq. (59).

B.2 Proof of Proposition 1

Proof. The preceding result shows that $\|\mathbf{x}\|^a J^{-b}$ is jointly convex for $a \geq 1$ and $0 \leq b \leq a - 1$. Composition with the linear maps $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i$ and $\mathbf{F} \mapsto \mathbf{F}\mathbf{d}_i^\perp$, followed by the positive weighted sum in Eq. (58), proves convexity of each feature in independent $(\mathbf{F}, J)$. Norms are unchanged by left rotations, proving objectivity.

For the reference derivative, evaluate the physical features with $J = \det \mathbf{F}$, holding all learned parameters fixed. At $\mathbf{E} = \mathbf{0}$, both directional invariants and J equal one, giving z_i^0 . Using $I_4(\mathbf{d}_i) = \mathbf{d}_i^T(\mathbf{I} + 2\mathbf{E})\mathbf{d}_i$ and $J^2 = (1 + 2E_{11})(1 + 2E_{22}) - 4E_{12}^2$ gives $\partial_{\mathbf{E}} I_4(\mathbf{d}_i) = 2\mathbf{d}_i \otimes \mathbf{d}_i$ and $\partial_{\mathbf{E}} J|_0 = \mathbf{I}$. The product rule therefore yields

$$\partial_{\mathbf{E}} \frac{I_4(\mathbf{d}_i)^{p_i}}{p_i J^{b_i}} \Big|_0 = \frac{1}{p_i} [2p_i \mathbf{d}_i \otimes \mathbf{d}_i - b_i \mathbf{I}] = 2\mathbf{d}_i \otimes \mathbf{d}_i - \frac{b_i}{p_i} \mathbf{I}. \quad (95)$$

The perpendicular contribution has the same form with $(\mathbf{d}_i, p_i, b_i)$ replaced by $(\mathbf{d}_i^\perp, q_i, c_i)$. Since $\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp = \mathbf{I}$, adding the two contributions gives

$$\partial_{\mathbf{E}} z_i|_0 = 2(\mathbf{d}_i \otimes \mathbf{d}_i + \mathbf{d}_i^\perp \otimes \mathbf{d}_i^\perp) - \left(\frac{b_i}{p_i} + \frac{c_i}{q_i} \right) \mathbf{I} = \rho_i \mathbf{I}. \quad (96)$$

In the work-conjugate engineering-strain convention of Eq. (2), this tensor gradient corresponds to $\partial_{\mathbf{e}} z_i|_0 = \rho_i(1, 1, 0)^T$, proving Eq. (61). $\square$

B.3 Proof of Proposition 2

Proof. Potential structure, objectivity, and stress symmetry (C1)–(C3). Every directional norm and J is unchanged by left proper rotations. The features can also be expressed through $\mathbf{d}_i^T \mathbf{C} \mathbf{d}_i$ and $\sqrt{\det \mathbf{C}}$. Hence the complete energy is objective, and differentiation with respect to symmetric strain gives a symmetric material stress. Differentiating this scalar energy supplies (C1) and the major symmetry of its strain tangent. Smoothness holds on $\text{GL}^+(2)$ because $J > 0$, nonzero material directions have nonzero deformed length, and both cores are C^2 . No right-rotation invariance is used in this argument.

Reference normalization (C4). Equations (68) and (70) establish zero reference energy and cancellation of the neural reference stress. The calculations for groups (iii)–(iv) in Section 4.5 show that the analytic growth terms have zero value and zero strain gradient at the reference. Thus the complete energy satisfies both reference conditions.

Polyconvexity (C5). Let $\mathbf{G} \in \mathbb{R}^{2 \times 2}$ and $j > 0$ be independent. Define $\mathcal{P}(\mathbf{G}, j)$ by the right-hand side of Eq. (66), replacing $\mathbf{z}$ by $\mathbf{z}(\mathbf{G}, j)$, J by j , and $\text{tr } \mathbf{C}$ by $\|\mathbf{G}\|_F^2$. For two such pairs $\mathbf{X}_1 = (\mathbf{G}_1, j_1)$, $\mathbf{X}_2 = (\mathbf{G}_2, j_2)$ and $0 \leq t \leq 1$,

$$\begin{aligned} H(\mathbf{z}(t\mathbf{X}_1 + (1-t)\mathbf{X}_2)) &\leq H(t\mathbf{z}(\mathbf{X}_1) + (1-t)\mathbf{z}(\mathbf{X}_2)) \\ &\leq tH(\mathbf{z}(\mathbf{X}_1)) + (1-t)H(\mathbf{z}(\mathbf{X}_2)). \end{aligned} \quad (97)$$

The first inequality uses feature convexity and core monotonicity; the second uses core convexity. The reference value is constant, and $-r(j-1)$ is affine for either sign of r . The remaining terms are convex in independent $(\mathbf{G}, j)$: their nonaffine parts are positive multiples of $-\log j$, $(j-1)^2$, and $\|\mathbf{G}\|_F^2$. Thus $\mathcal{P}$ is convex and $W(\mathbf{F}) = \mathcal{P}(\mathbf{F}, \det \mathbf{F})$, establishing Eq. (52).

Growth (C6). Convexity of H gives its supporting-plane inequality at $\mathbf{z}^0$:

$$H(\mathbf{z}) \geq H(\mathbf{z}^0) + \sum_i h_i(z_i - z_i^0) \geq H(\mathbf{z}^0) - \sum_i h_i z_i^0. \quad (98)$$

The second inequality uses $h_i, z_i \geq 0$. The resulting lower bound is independent of deformation, so the neural contribution cannot cancel the divergent positive growth terms. As $J \rightarrow 0^+$, the logarithmic barrier diverges and the affine determinant correction remains bounded. As $J \rightarrow \infty$, the positive quadratic determinant term dominates all potentially negative linear and logarithmic terms. For J confined to any compact interval in $(0, \infty)$, the positive term $\varepsilon \|\mathbf{F}\|_F^2/2$ diverges as $\|\mathbf{F}\|_F \rightarrow \infty$ while the determinant terms stay bounded. These establish the three stated growth limits. $\square$

C Sufficient lower bound for nonnegative energy

Write $t = \|\mathbf{F}\mathbf{d}_i\|^2$, $u = \|\mathbf{F}\mathbf{d}_i^\perp\|^2$. Hadamard's determinant inequality gives $tu \geq J^2$. Since each feature is increasing in t, u , its minimum at fixed J occurs on $tu = J^2$. Minimizing

$$\frac{t^{p_i}}{p_i J^{b_i}} + \frac{J^{2q_i - c_i} t^{-q_i}}{q_i} \quad (99)$$

gives $t^{p_i + q_i} = J^{2q_i + b_i - c_i}$ and hence $z_i \geq k_i J^{\eta_i}$. This is a separate lower bound for each feature; simultaneous attainability for all directions is not assumed. It therefore remains valid after summing the supporting-plane inequalities.

For Eq. (74),

$$J^2 g''(J) = \alpha + \varepsilon + \beta J^2 + \sum_i c_i^* J^{\eta_i}, \quad c_i^* = h_i k_i \eta_i (\eta_i - 1). \quad (100)$$

Only terms with $0 < \eta_i < 1$ can have negative coefficients. If $c_- = \sum_{c_i^* < 0} |c_i^*|$, the two inequalities

$$\alpha + \varepsilon + \sum_{\eta_i < 0} c_i^* - c_- \geq 0, \quad (101)$$

$$\beta + \sum_{\eta_i \geq 1} c_i^* - c_- \geq 0 \quad (102)$$

are sufficient on $0 < J \leq 1$ and $J \geq 1$, respectively. For the first region, $J^{\eta_i} \leq 1$ for negative-coefficient terms and $J^{\eta_i} \geq 1$ when $\eta_i < 0$. For the second region, divide $J^2 g''$ by J : all negative-coefficient powers then satisfy $J^{\eta_i - 1} \leq 1$, while $J^{\eta_i - 1} \geq 1$ for $\eta_i \geq 1$ and $\beta J \geq \beta$. Dropping the remaining nonnegative contributions gives the stated bounds. Their failure is inconclusive.

The selected parameters require a sharper check. The audit partitions $[10^{-4}, 10^4]$ into 4000 logarithmically spaced intervals. On each interval $[J_L, J_R]$, a positive power term is bounded below by the smaller endpoint value, whereas a negative-coefficient term is bounded below by $c_i^* J_R^{\eta_i}$, since its exponent lies in $(0, 1)$. The constant and quadratic terms are bounded below by $\alpha + \varepsilon$ and βJ_L^2 . Below the finite interval, the nonnegative negative-exponent terms increase and the magnitudes of the negative-coefficient terms decrease toward $J = 0$. Above it, factoring out J^{η_-} with $\eta_- = \max_{c_i^* < 0} \eta_i < 1$ yields a lower bound whose retained positive powers increase. These endpoint and tail inequalities cover every $J > 0$, not only the grid points. Positive margins establish the sufficient analytical argument for the saved parameters to the accuracy of its floating-point evaluation; random energy samples alone would not establish it.

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